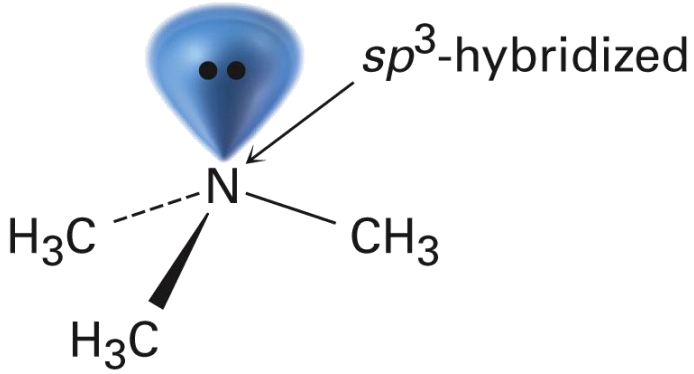
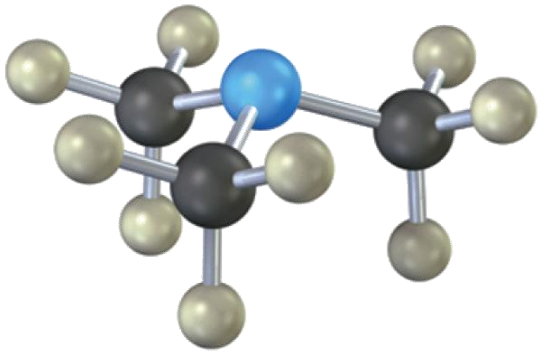
Chapter 24
Amines and Heterocycles

Structure and Properties of Amines

- Bonding in alkylamines is similar to that in ammonia
  - N is sp^3-hybridized
  - C–N–C bond angles are close to 109° tetrahedral value

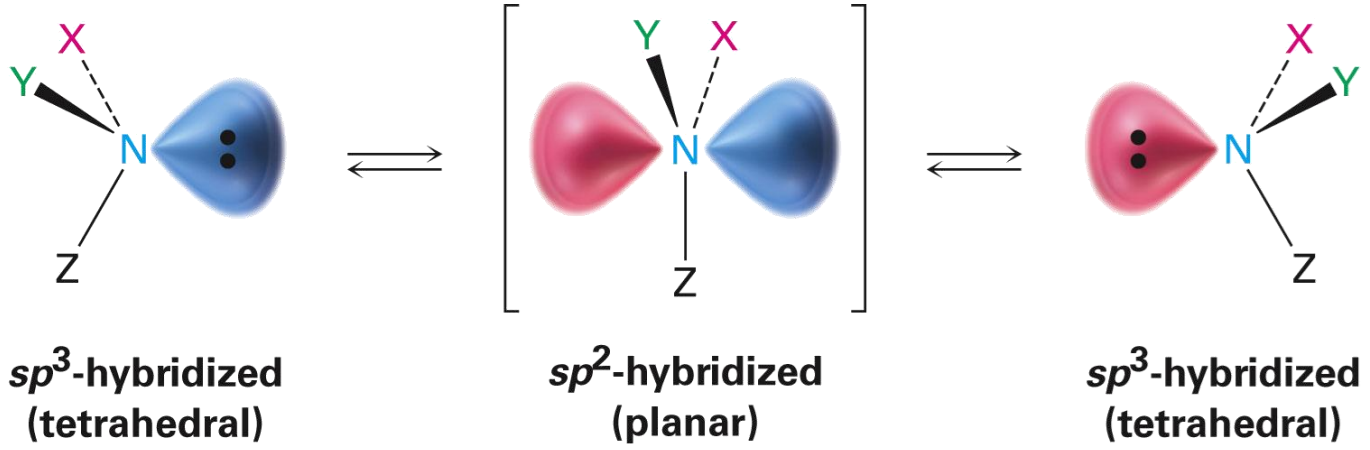


Trimethylamine



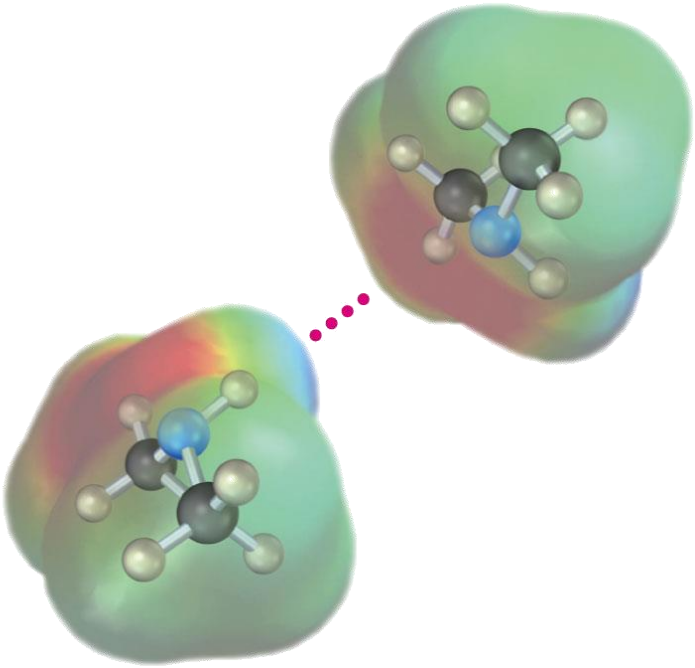
Structure and Properties of Amines

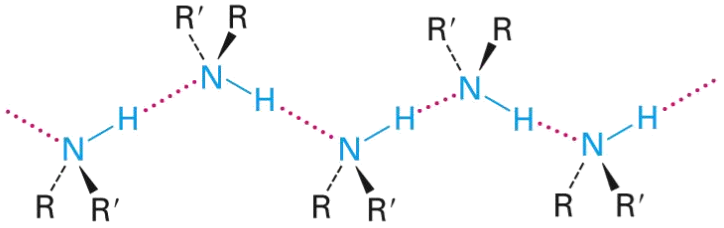
- An amine with three different substituents on nitrogen is chiral
  - The two enantiomeric forms rapidly interconvert by a pyramidal inversion



Structure and Properties of Amines

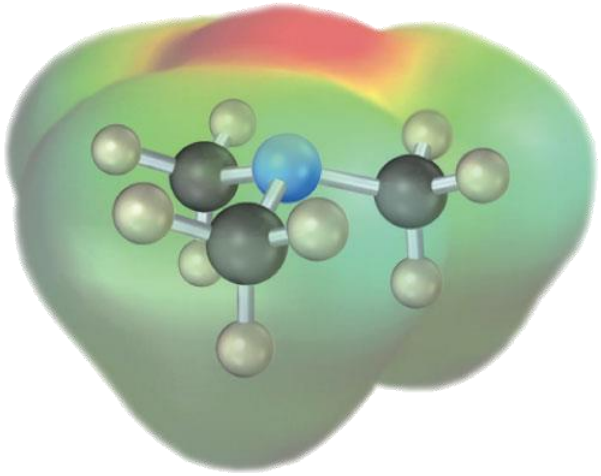
- Amines with fewer than five carbon atoms are water-soluble
- Primary and secondary amines form hydrogen bonds, increasing their boiling points





Basicity of Amines

- Lone pair of electrons on nitrogen makes amines basic and nucleophilic
  - React with acids to form acid-base salts and they react with electrophiles



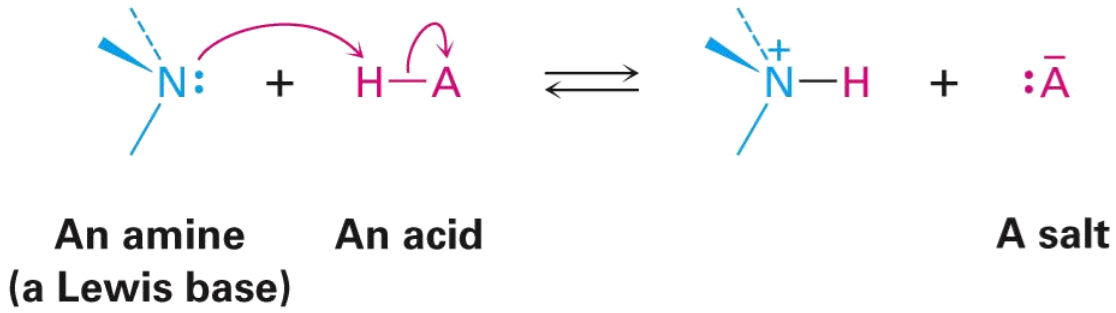
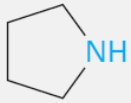
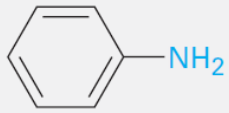
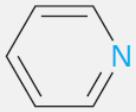
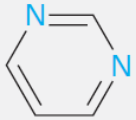
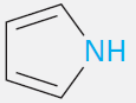
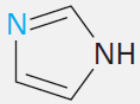


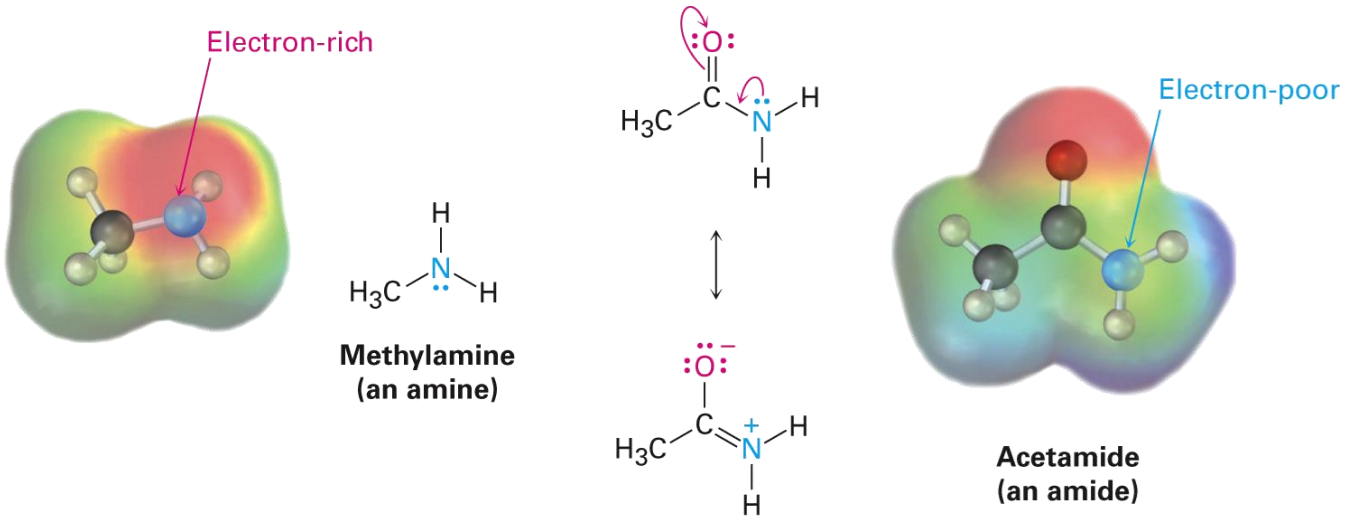
Table 24.1 - Basicity of Some Common Amines

Name	Structure	pK_a of ammonium ion
Ammonia	NH_3	9.26
Primary alkylamine		
Methylamine	CH_3NH_2	10.64
Ethylamine	$CH_3CH_2NH_2$	10.75
Secondary alkylamine		
Diethylamine	$(CH_3CH_2)_2NH$	10.98
Pyrrolidine		11.27
Tertiary alkylamine		
Triethylamine	$(CH_3CH_2)_3N$	10.76
Arylamine		
Aniline		4.63

Name	Structure	pK_a of ammonium ion
Heterocyclic amine		
Pyridine		5.25
Pyrimidine		1.3
Pyrrole		0.4
Imidazole		6.95

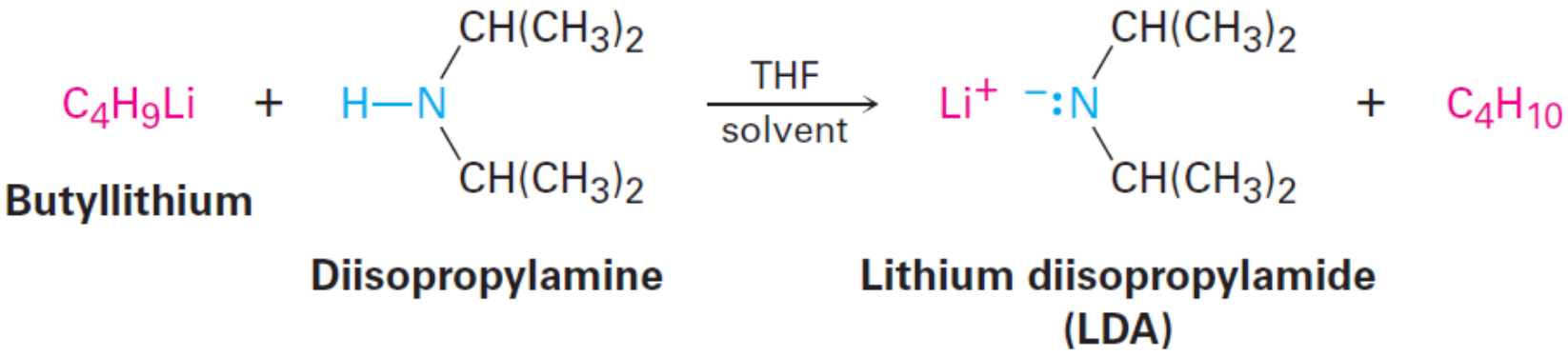
Basicity of Amines

- Amides ($RCONH_2$) are nonbasic, in contrast with amines
- Amides are stabilized by delocalization of the nitrogen lone-pair electrons
- Amides are more stable than amines
  - Stability is lost when protonated



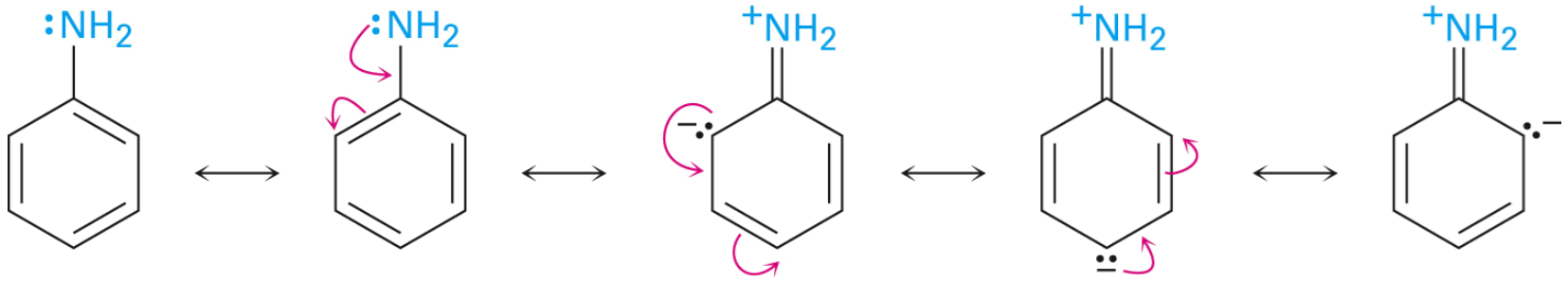
Basicity of Amines

- Primary and secondary amines can act as very weak acids
- N-H proton can be removed by a sufficiently strong base



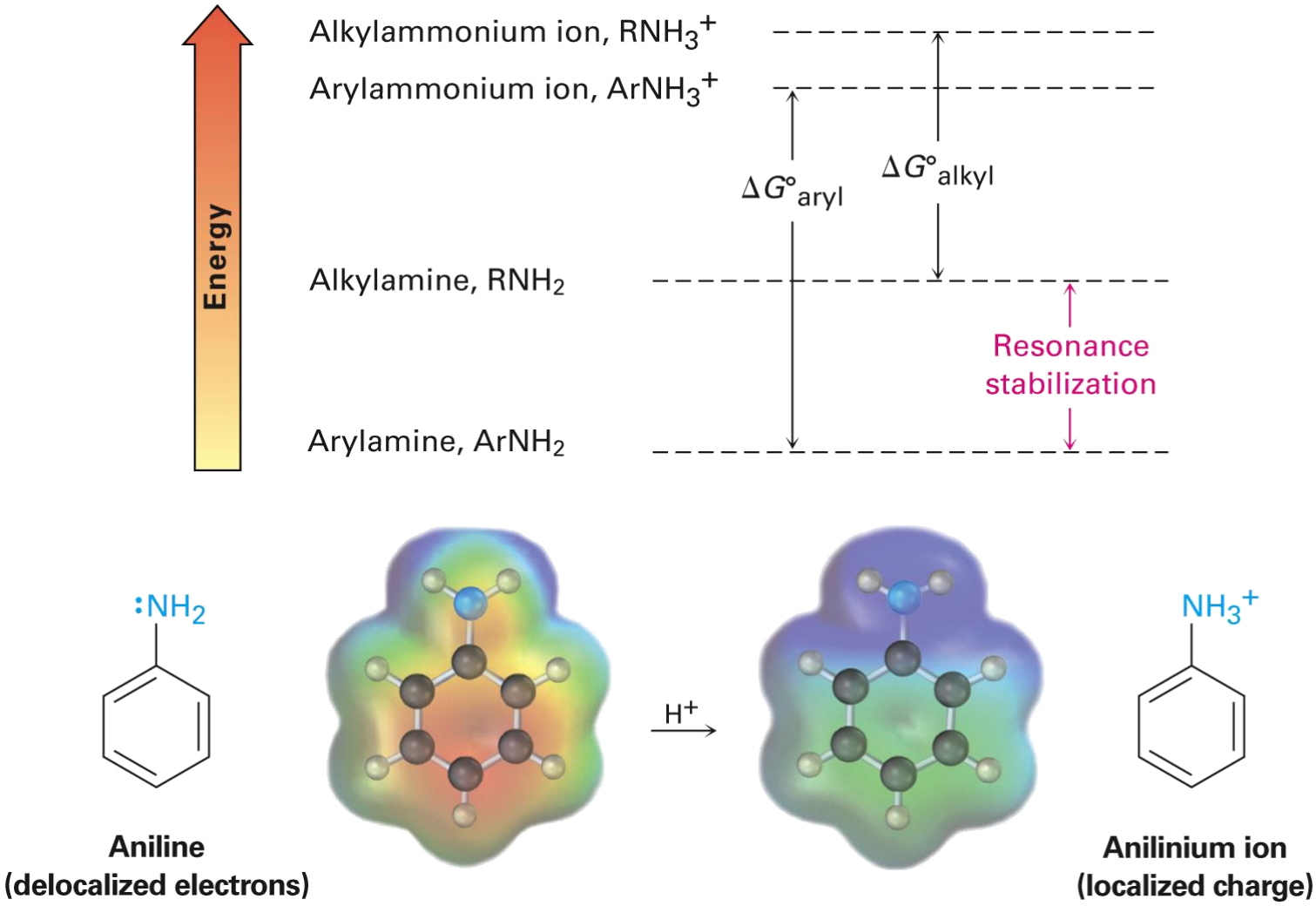
Basicity of Arylamines

- Arylamines are less basic than alkylamines
  - The N lone-pair electrons in arylamines are delocalized by interaction with the aromatic ring's π electron system
  - Are less able to accept H^+



- Energy difference between protonated and nonprotonated forms is higher for arylamines

Figure 24.3 - Electrostatic Potential Maps



Basicity of Arylamines

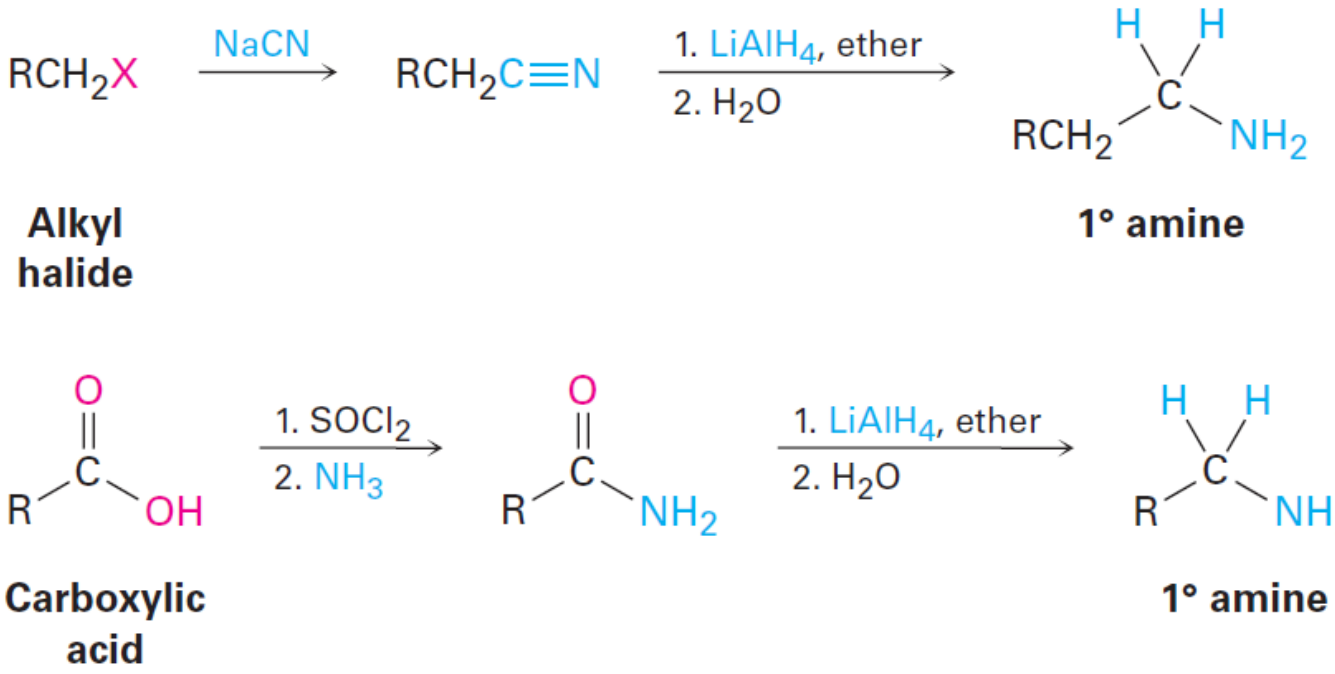
- Substituted arylamines can be either more basic or less basic than aniline
- Electron-donating substituents increase the basicity of the corresponding arylamine
- Electron-withdrawing substituents decrease arylamine basicity



	Substituent, Y	pK_a	
Stronger base	$-NH_2$	6.15	Activating groups
	$-OCH_3$	5.34	
	$-CH_3$	5.08	
	$-H$	4.63	
	$-Cl$	3.98	Deactivating groups
	$-Br$	3.86	
	$-CN$	1.74	
Weaker base	$-NO_2$	1.00	

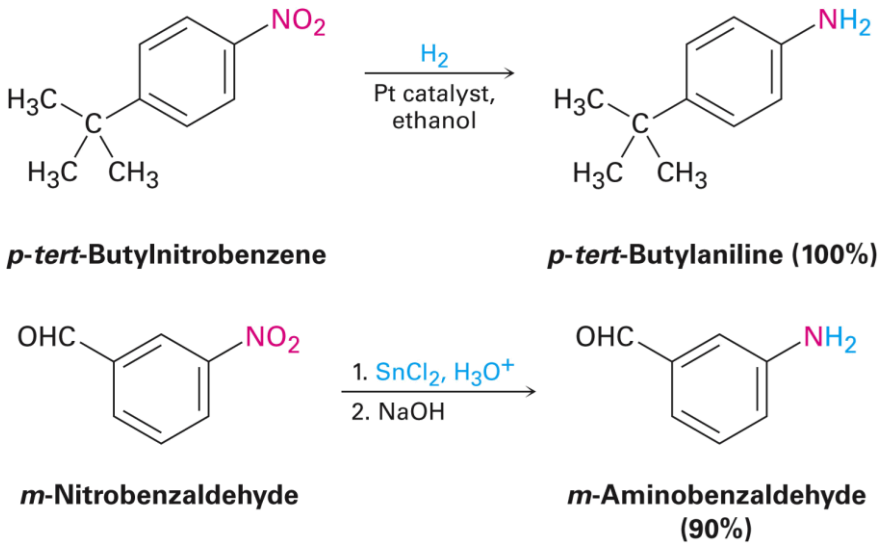
Synthesis of Amines

- Reduction of nitriles, amides, and nitro compounds
  - Amines can be prepared by reduction of nitriles and amides with $LiAlH_4$



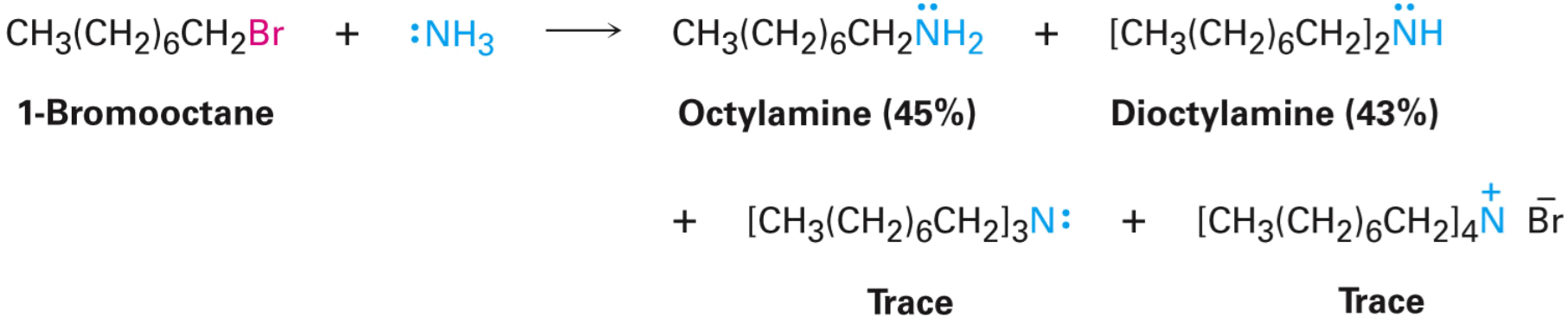
Synthesis of Amines

- Arylamines are prepared from nitration of an aromatic compound and reduction of the nitro group
- Reduction by catalytic hydrogenation over platinum is suitable if no other groups can be reduced
- Iron, zinc, tin, and tin(II) chloride are effective in acidic solution



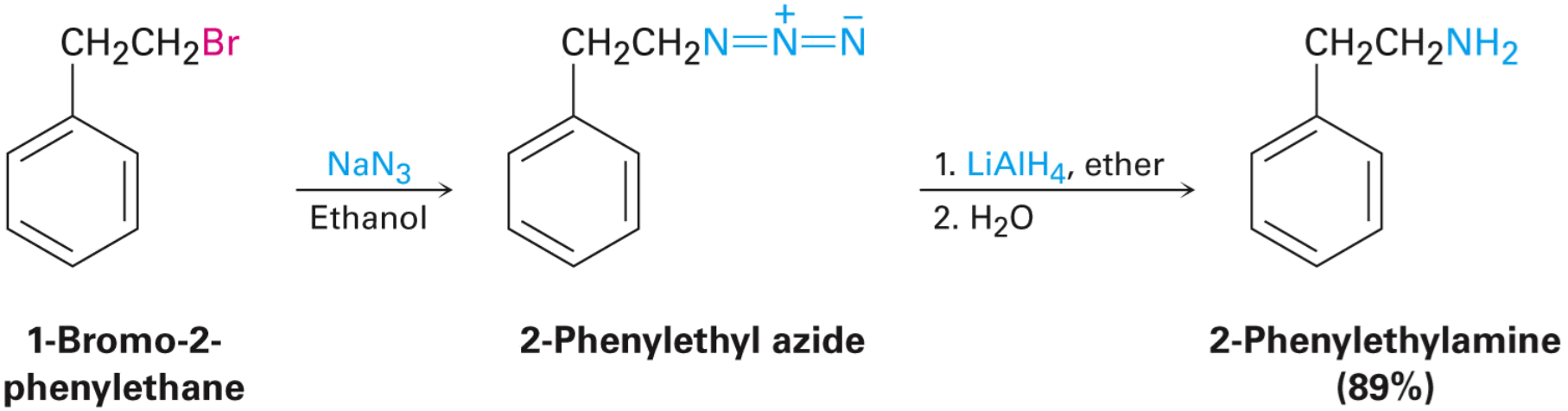
S_N2 Reactions of Alkyl Halides

- Primary, secondary, and tertiary amines all have similar reactivity
  - Initially formed monoalkylated substance undergoes further reaction to yield a mixture of products
- Secondary and tertiary amines undergo further alkylation



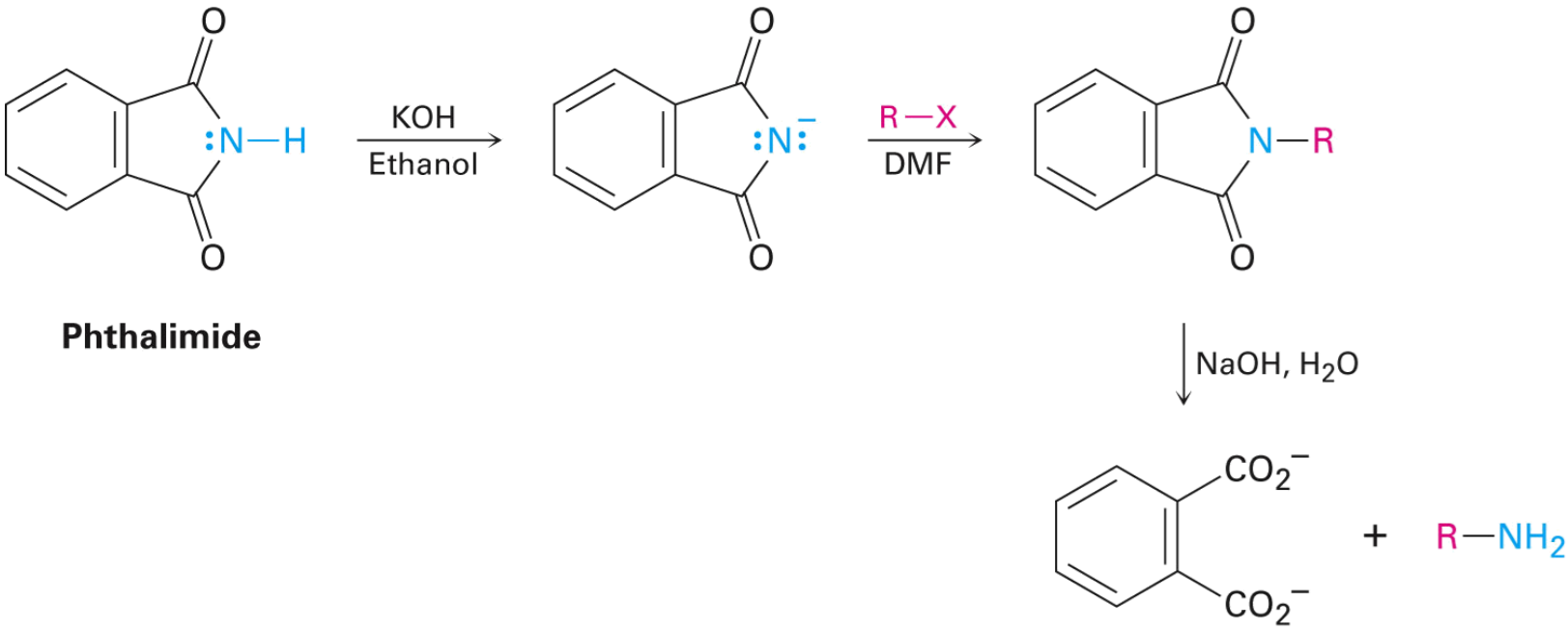
S_N2 Reactions of Alkyl Halides

- Azide ion, N_3^- displaces a halide ion from a primary or secondary alkyl halide to give an alkyl azide
- Alkyl azides are not nucleophilic
- Reduction gives the primary amine



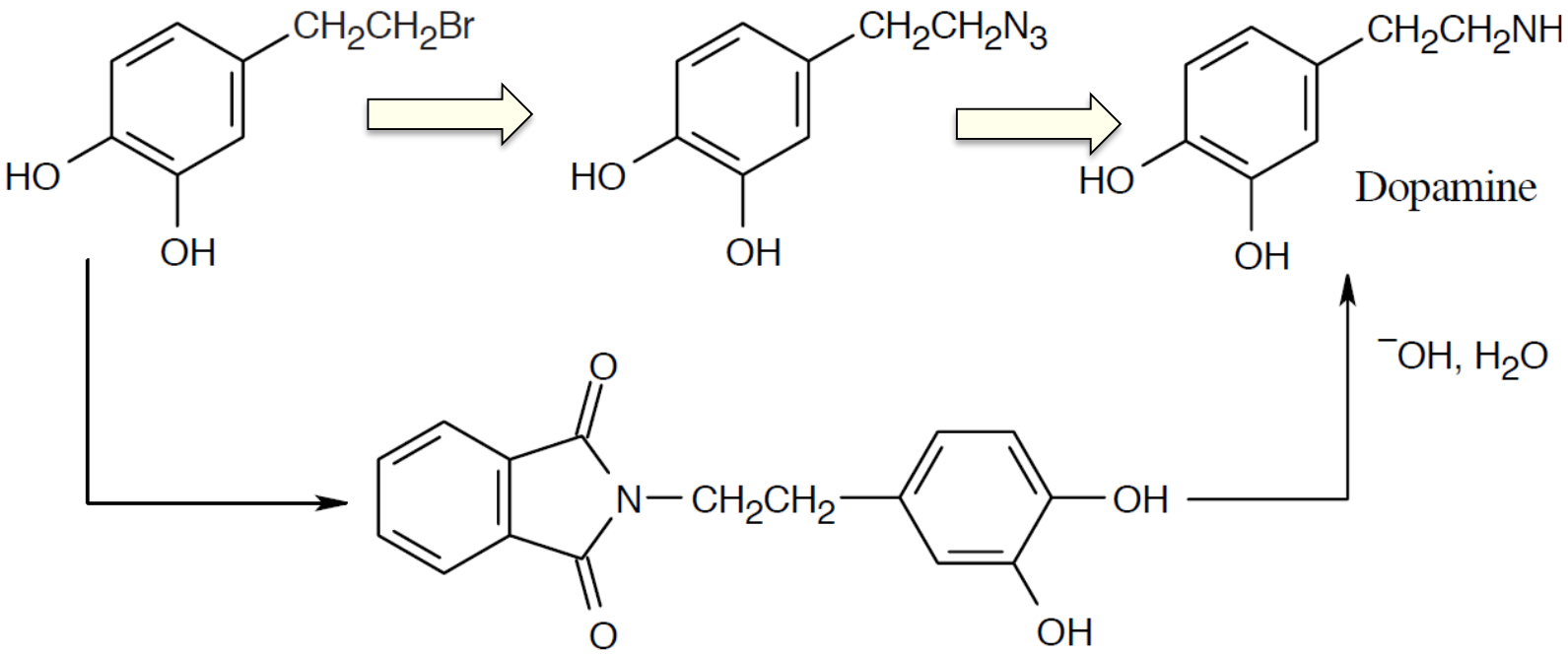
S_N2 Reactions of Alkyl Halides

- **Gabriel amine synthesis**: A phthalimide alkylation for preparing a primary amine from an alkyl halide
- N-H in **imides (–CONHCO–)** can be removed by KOH followed by alkylation and hydrolysis



Phthalimide

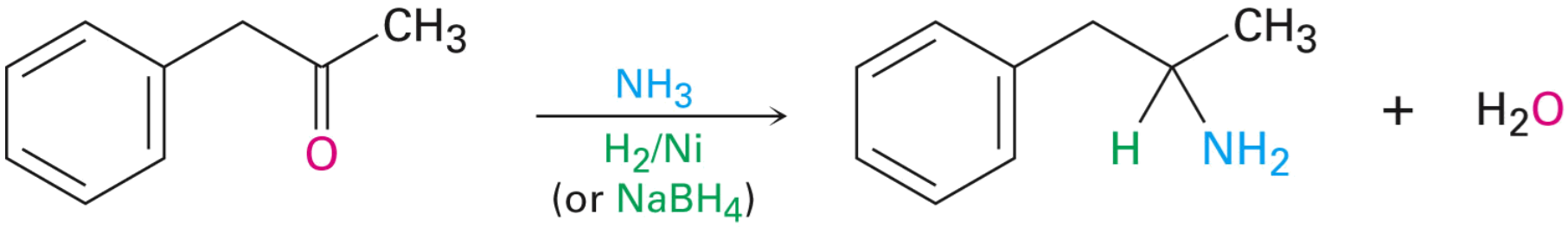
Worked Example



- Upper reaction is azide synthesis
- Lower reaction is Gabriel synthesis

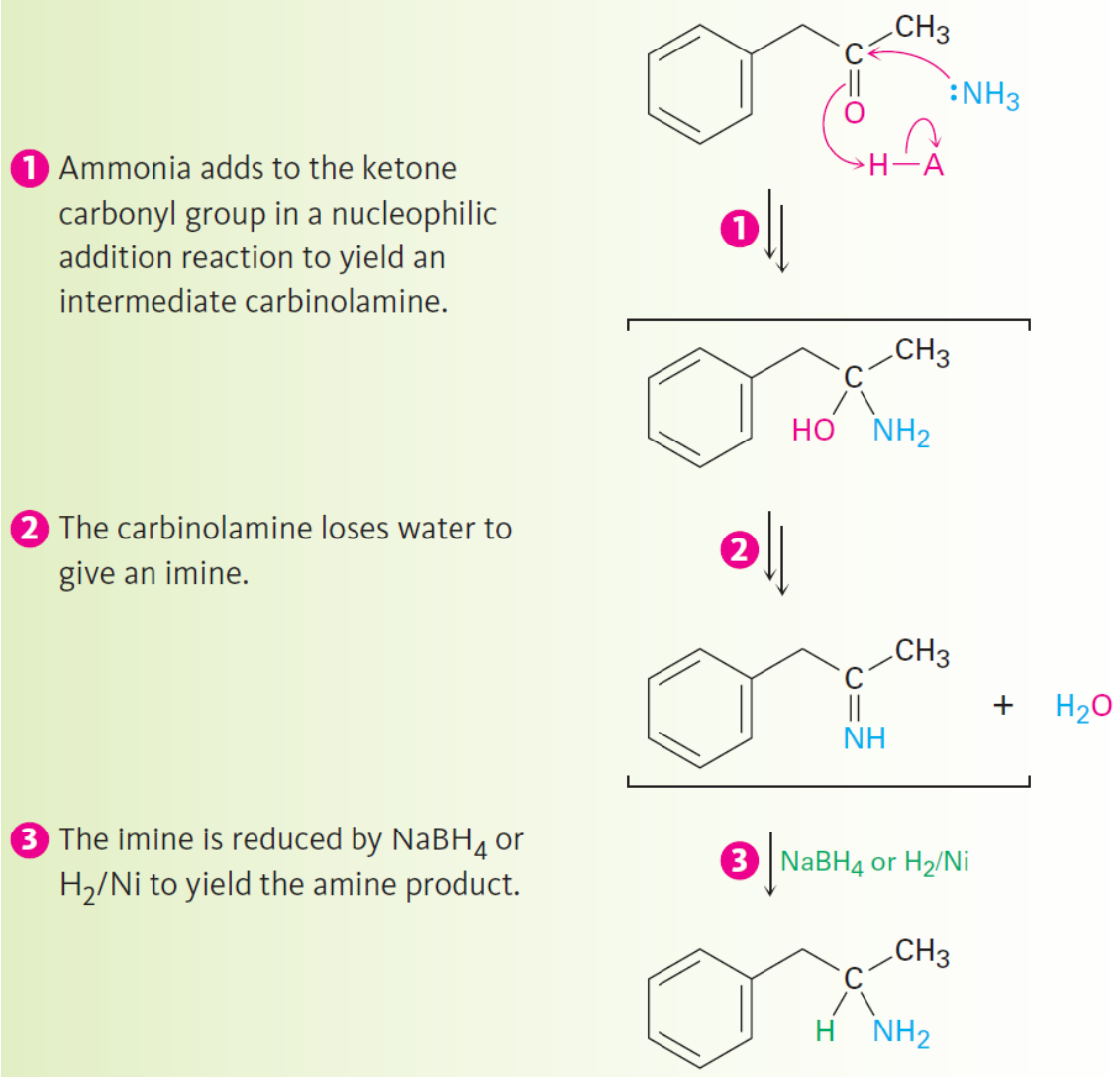
Reductive Amination of Aldehydes and Ketones

- **Reductive amination**: Treatment of an aldehyde or ketone with ammonia or an amine in the presence of a reducing agent



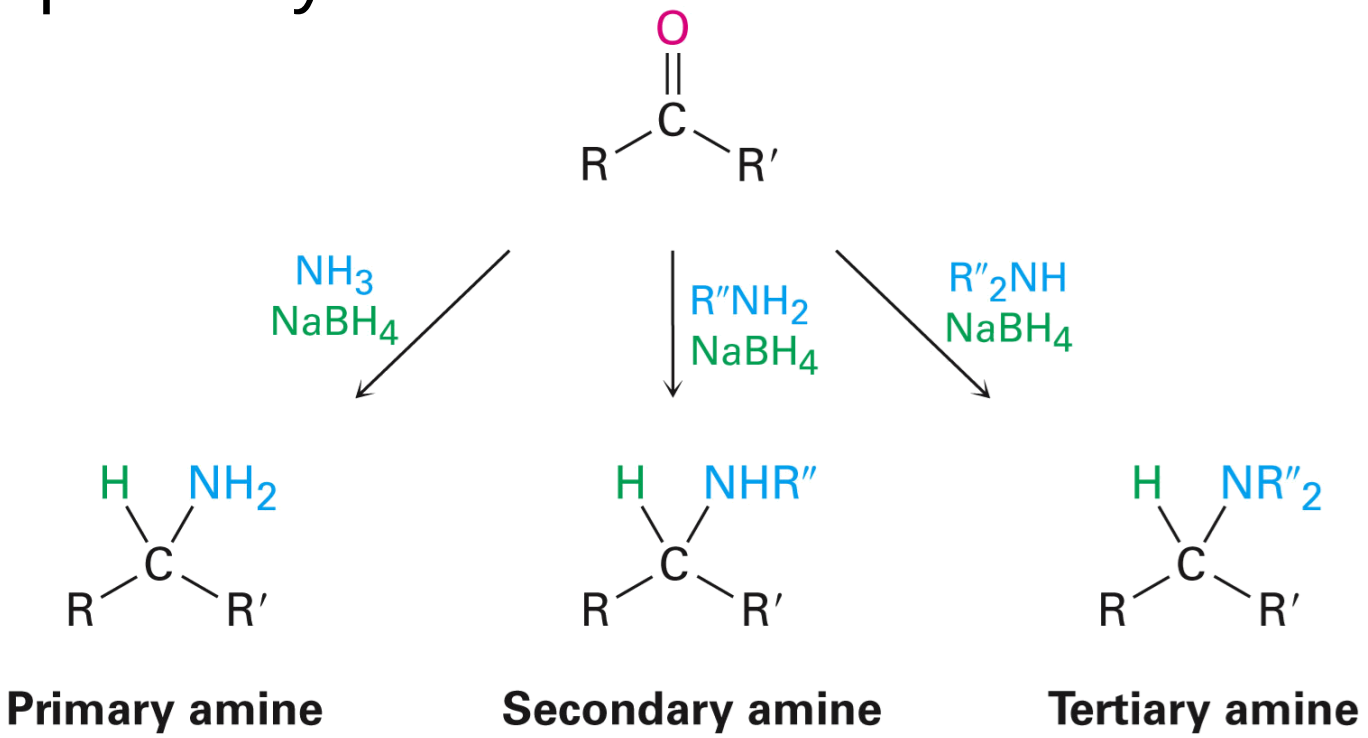
Phenyl-2-propanone **Amphetamine**

Figure 24.4 - Mechanism



Reductive Amination

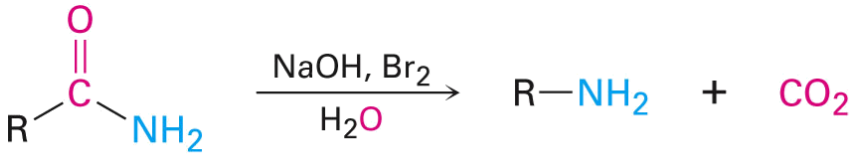
- Ammonia, primary amines, and secondary amines through reductive amination reaction yield primary, secondary, and tertiary amines, respectively



Hofmann and Curtius Rearrangements

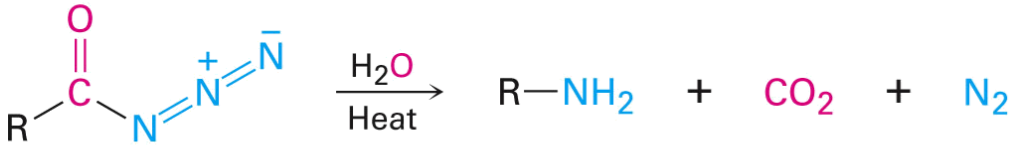
- Carboxylic acid derivatives can be converted into primary amines with loss of one carbon atom by both the **Hofmann rearrangement** and the **Curtius rearrangement**

Hofmann rearrangement



An amide

Curtius rearrangement



An acyl azide

Figure 24.5 - Mechanism

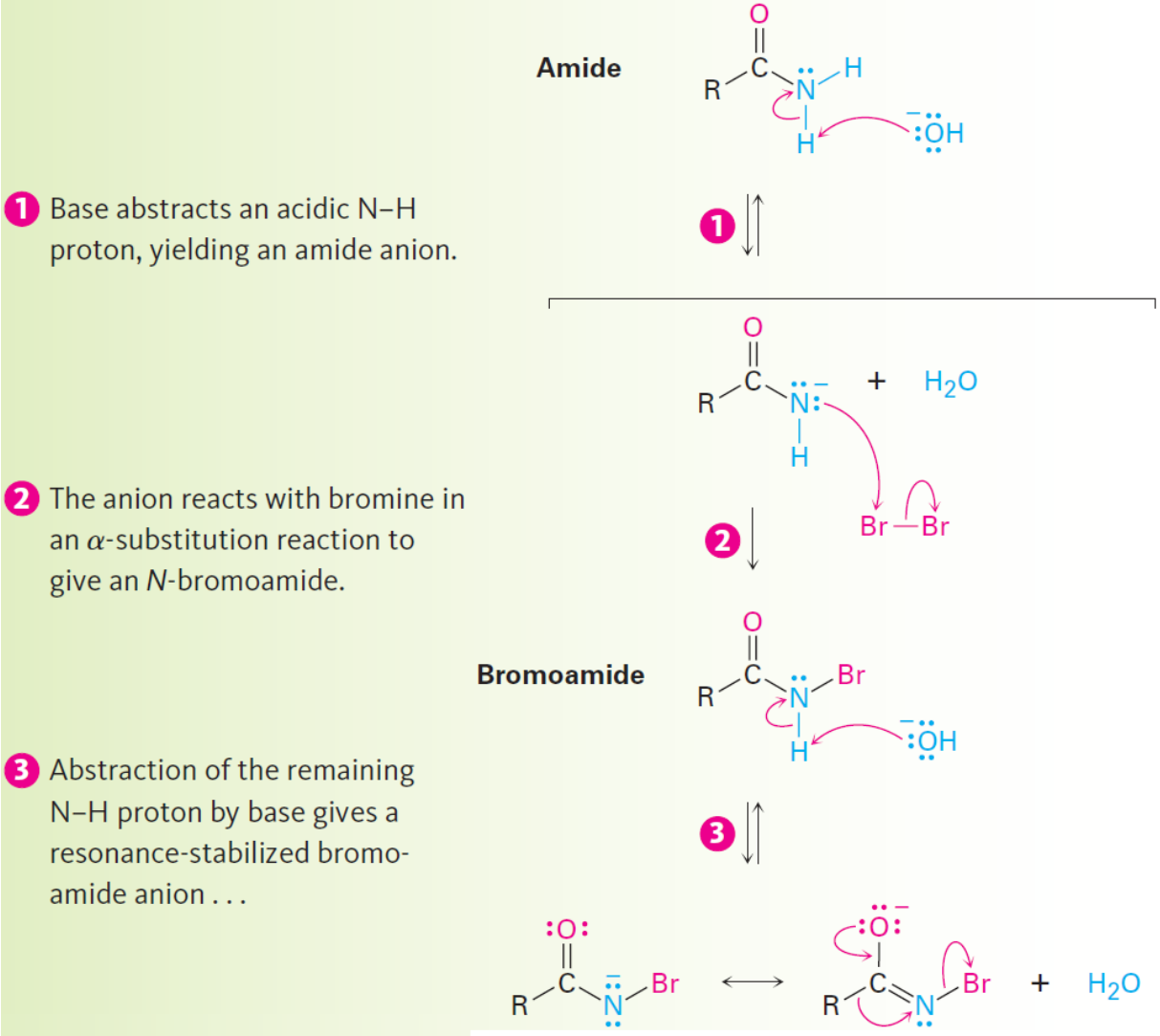
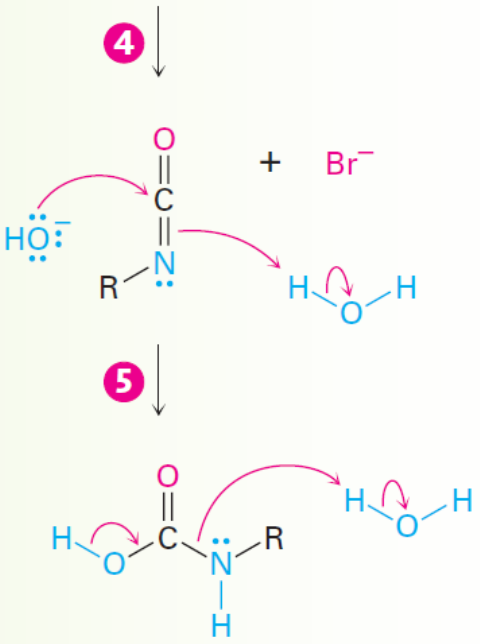


Figure 24.5 - Mechanism

4 . . . which rearranges when the R group attached to the carbonyl carbon migrates to nitrogen at the same time the bromide ion leaves.

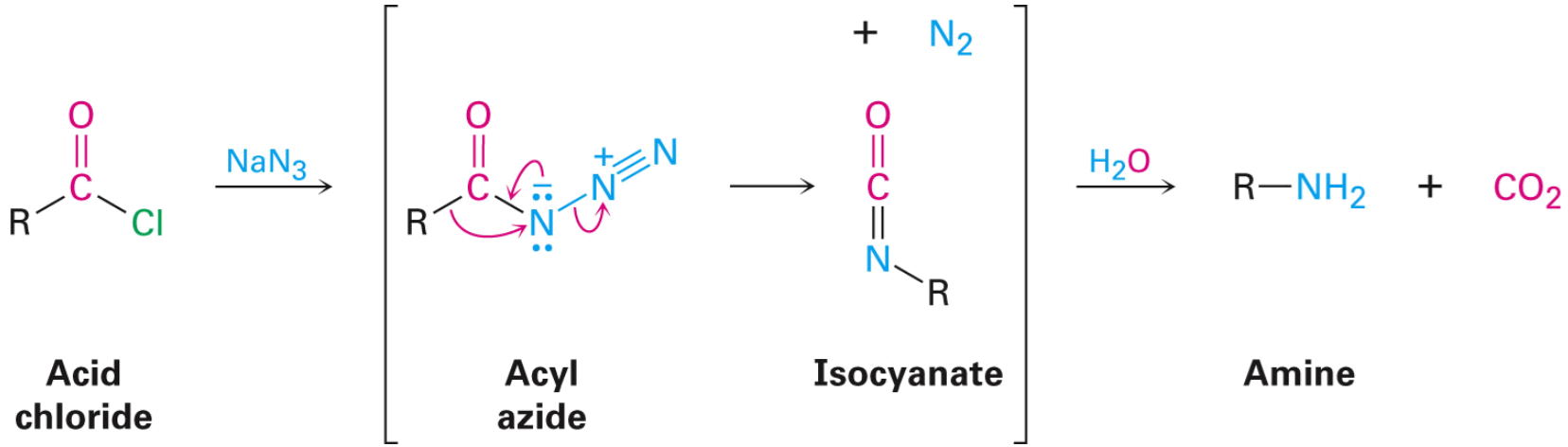
5 The isocyanate formed on rearrangement adds water in a nucleophilic addition step to yield a carbamic acid.

6 The carbamic acid spontaneously loses CO_2 to give an amine.



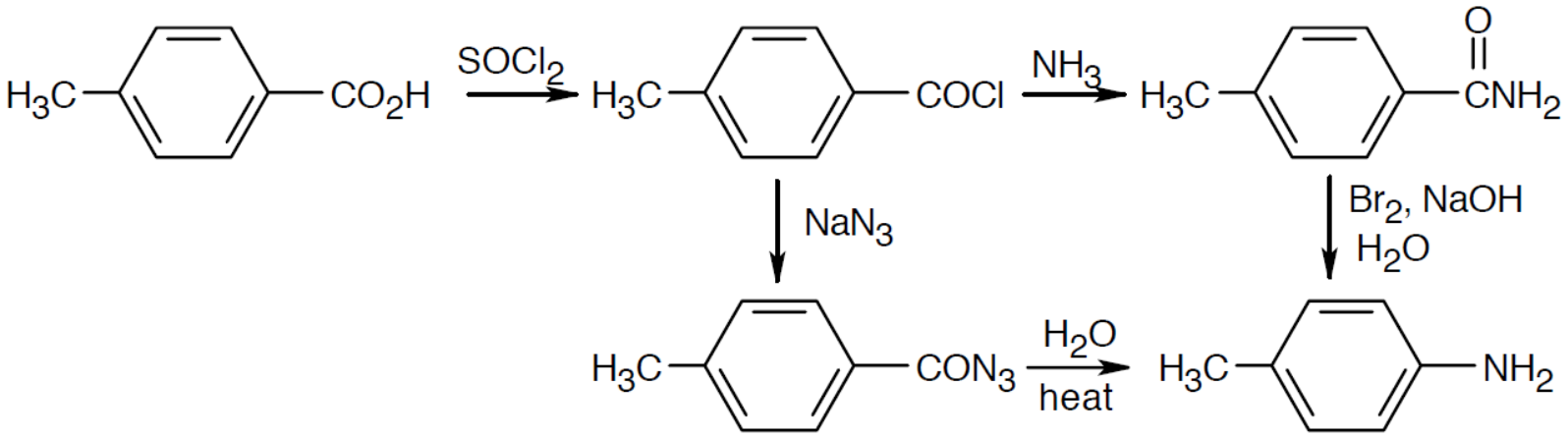
Curtius Rearrangement

- Heating an acyl azide prepared from an acid chloride
- Migration of –R from C=O to the neighboring nitrogen with simultaneous loss of a leaving group



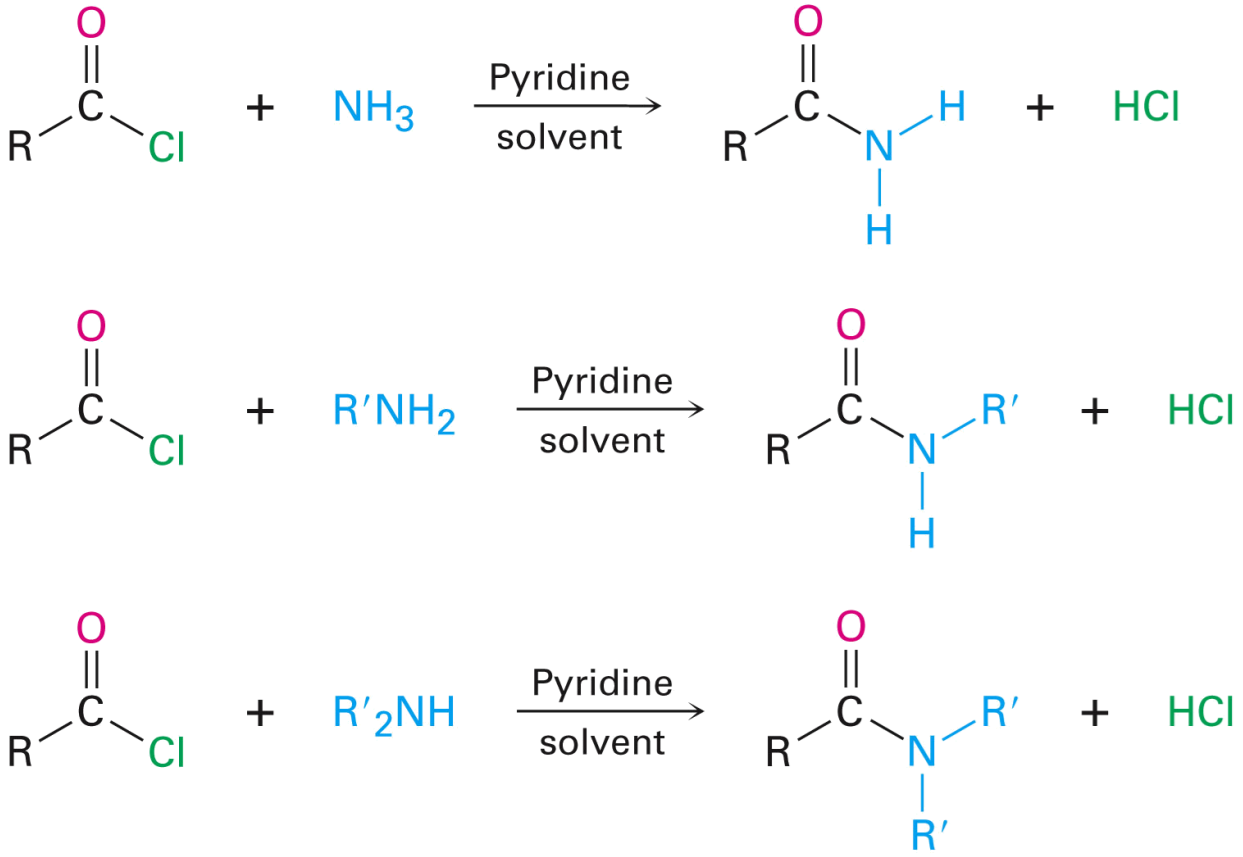
Worked Example

- Solution:
  - The precursor is an acid chloride, which is treated with NaN_3, then with H_2O and heat



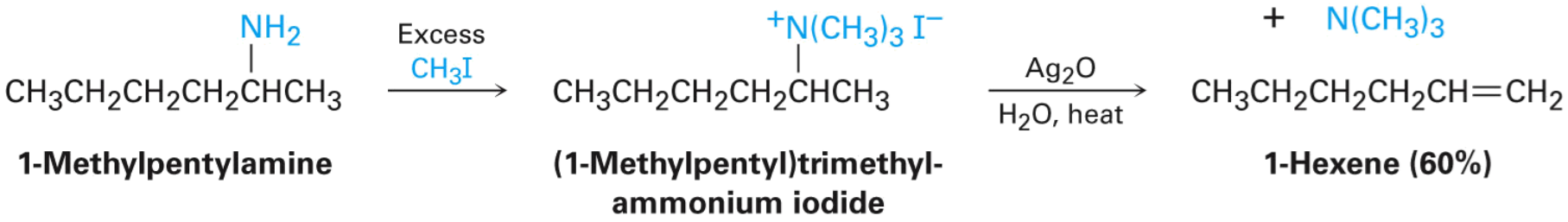
Reactions of Amines

- Primary and secondary amines can also be acylated by nucleophilic acyl substitution reaction



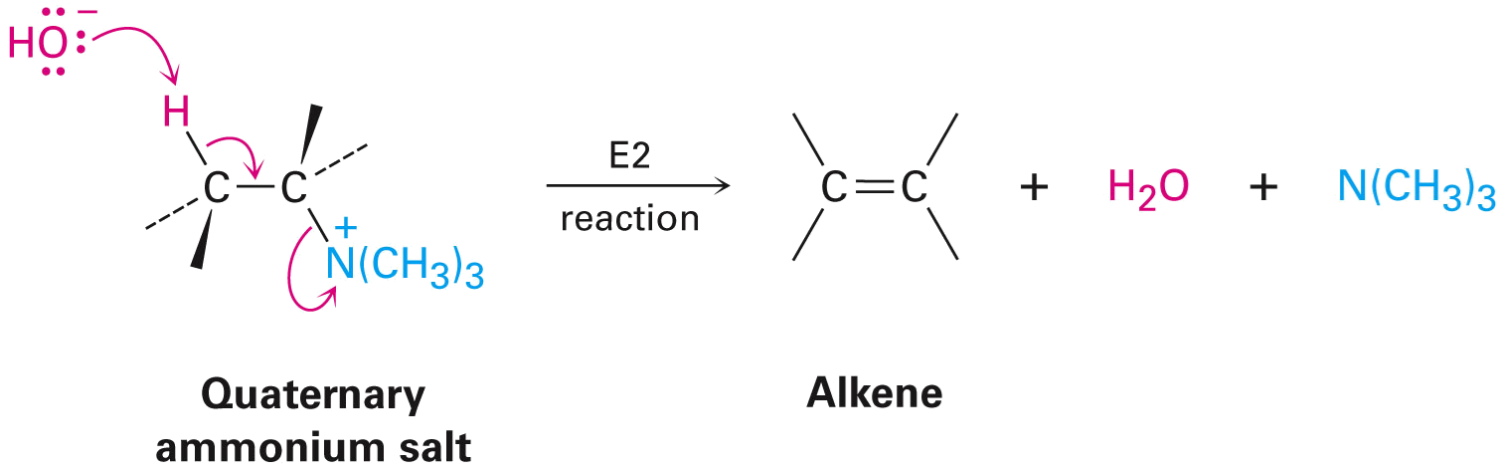
Hofmann Elimination

- Converts amines into alkenes
- NH_2^- is very a poor leaving group; it is converted to an alkylammonium ion, which is a good leaving group
- An amine is completely methylated by reaction with an excess amount of iodomethane to produce the corresponding quaternary ammonium salt

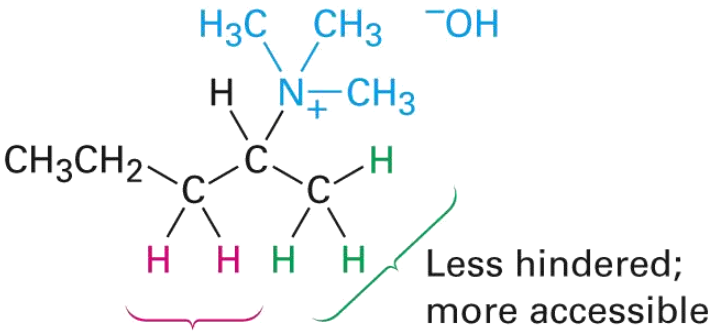


Hofmann Elimination

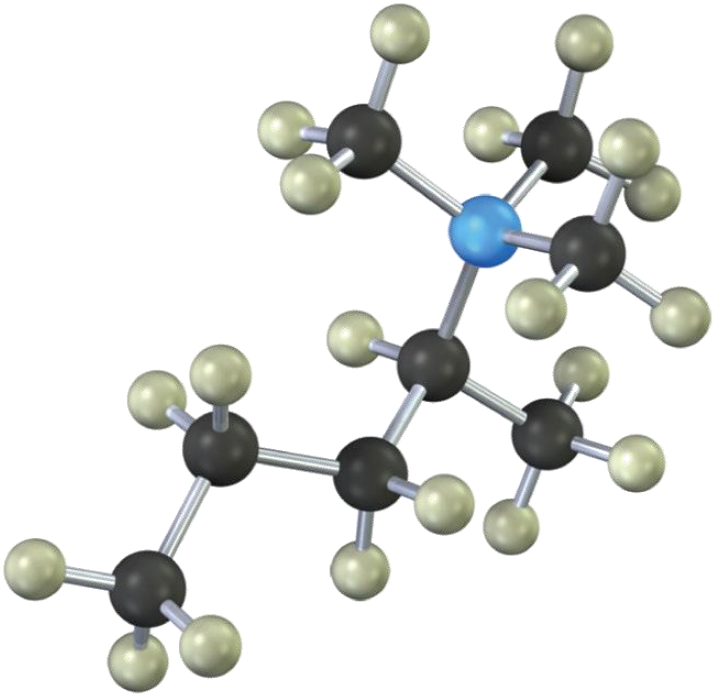
- Silver oxide is used for the elimination step
- Exchanges hydroxide ion for iodide ion in the quaternary ammonium salt, thus providing the base necessary to cause elimination



Orientation in Hofmann Elimination



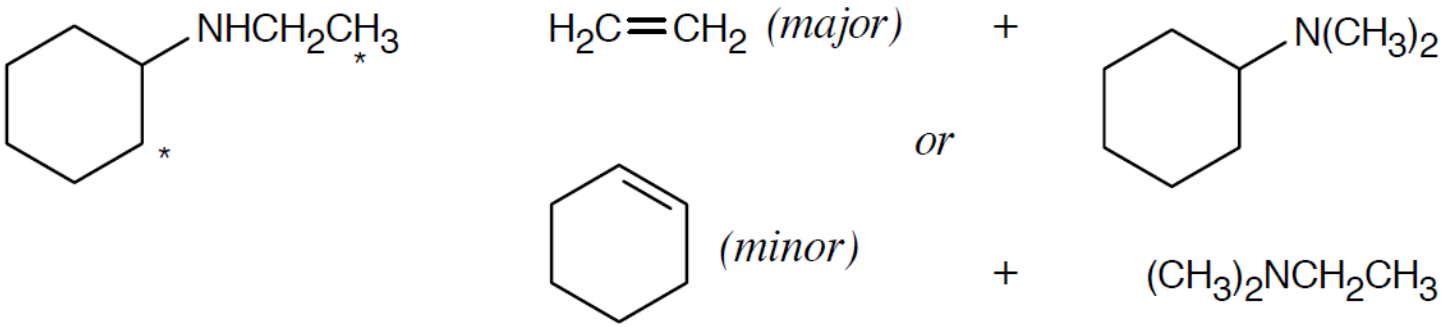
(1-Methylbutyl)trimethylammonium hydroxide



$CH_3CH_2CH_2CH{=}CH_2$ + $CH_3CH_2CH{=}CHCH_3$

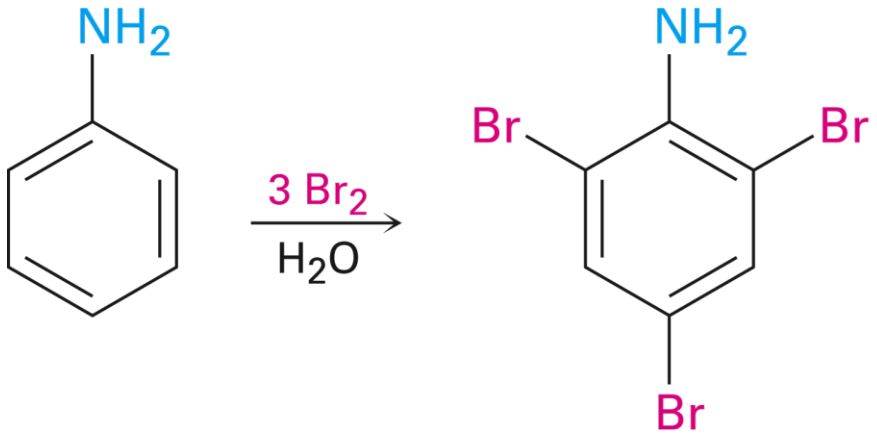
1-Pentene (94%) **2-Pentene (6%)**

Worked Example



Reactions of Arylamines

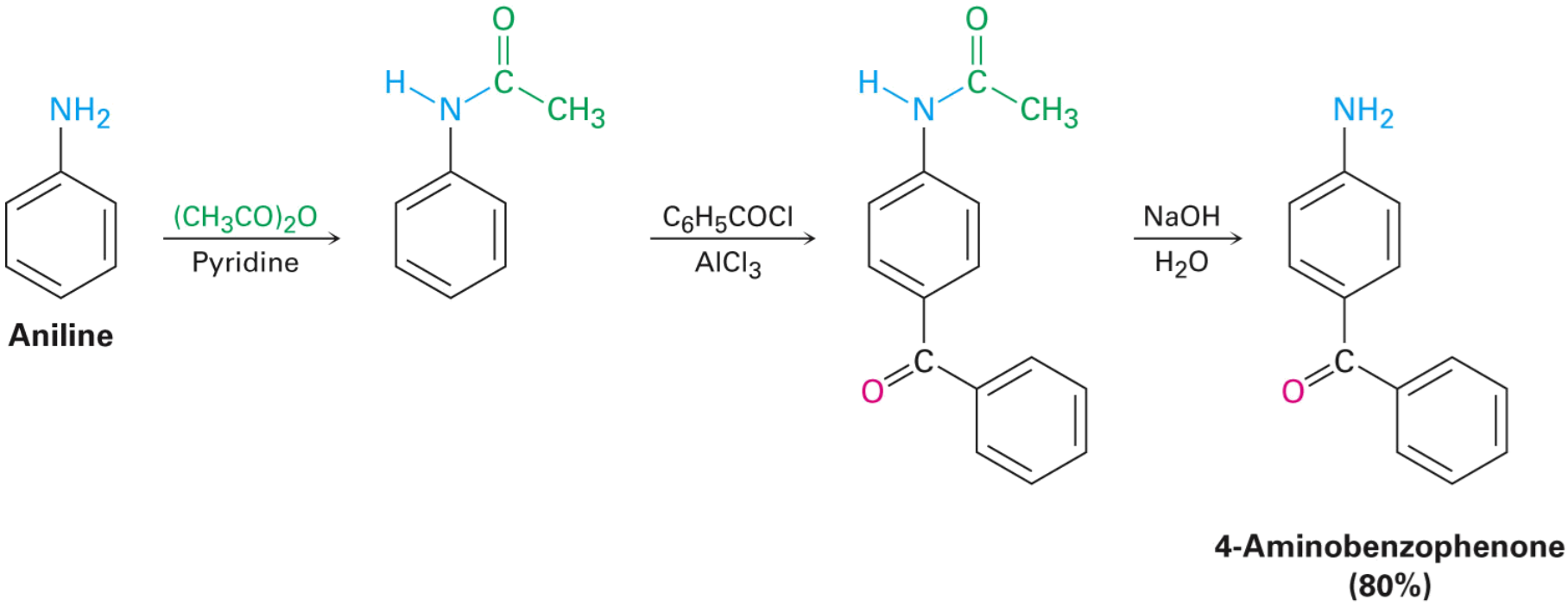
- Electrophilic aromatic substitution
  - Amino substituents are strongly activating and ortho- and para-directing groups in electrophilic aromatic substitution reactions
  - Reactions are controlled by conversion to amide



Aniline **2,4,6-Tribromoaniline (100%)**

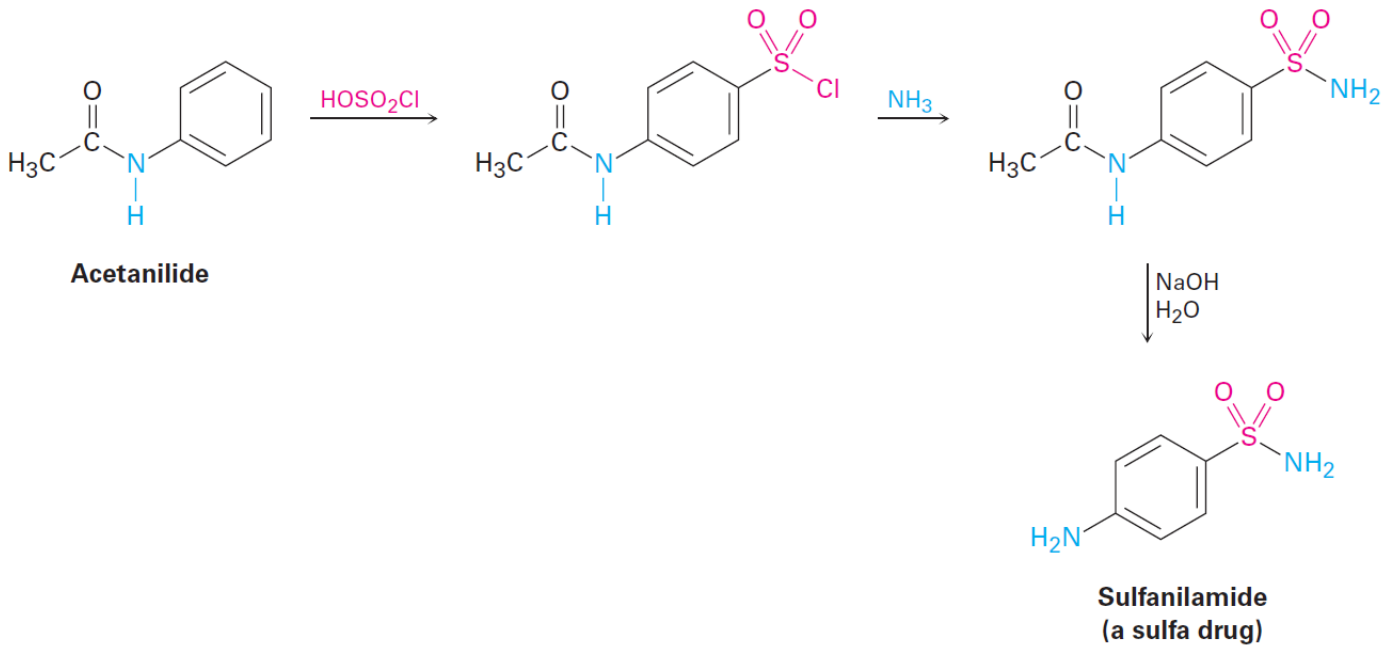
Reactions of Arylamines

- The amino group forms acid-base complex with the $AlCl_3$ catalyst preventing further reaction
- Therefore, we use the corresponding amide



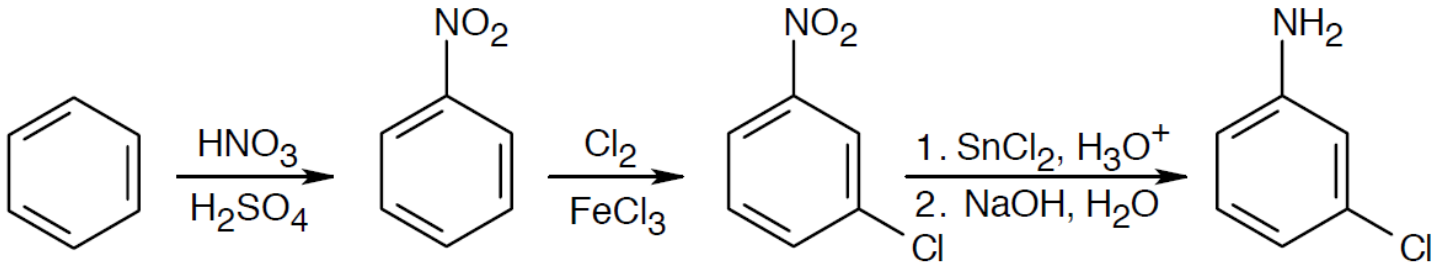
Reactions of Arylamines

- Modulating the reactivity of an amino-substituted benzene allows many kinds of electrophilic aromatic substitutions to be carried out
- Sulfa drugs were among the first pharmaceutical agents to be used clinically against bacterial infection



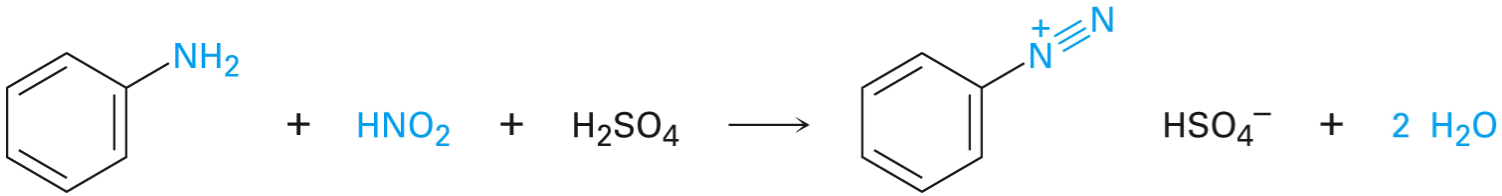
Worked Example

- Propose syntheses of *m*-Chloroaniline from benzene

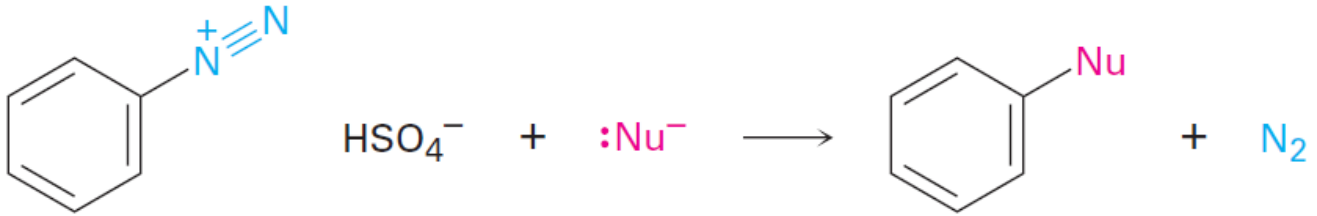


Reactions of Arylamines

- Diazonium salts: The Sandmeyer reaction
  - Primary arylamines react with HNO_2 yielding stable arenediazonium salts
    - Corresponding alkanediazonium can not be isolated

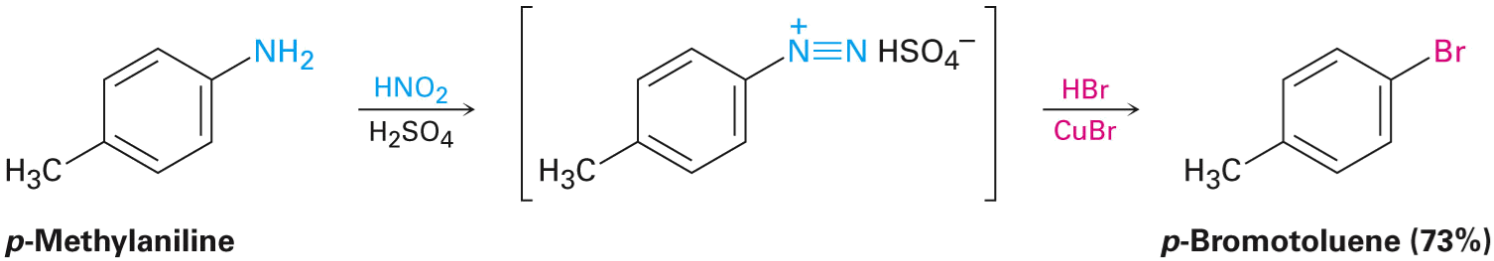


  - Arenediazonium salts are useful because the diazonio group can be replaced by a nucleophile in a substitution reaction

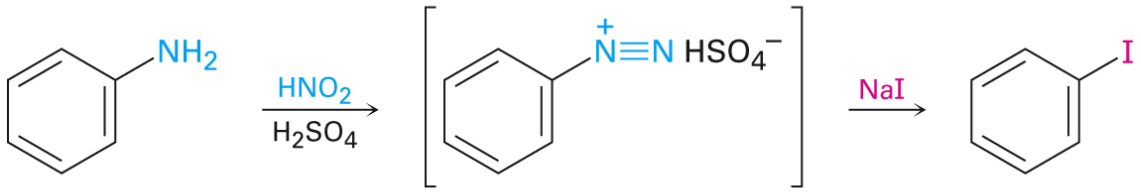


Reactions of Arylamines

- **Sandmeyer reaction**: Reaction of an arenediazonium salt with the corresponding copper(I) halide to yield aryl chlorides and bromides
- Aryl iodides can be prepared by direct reaction with NaI



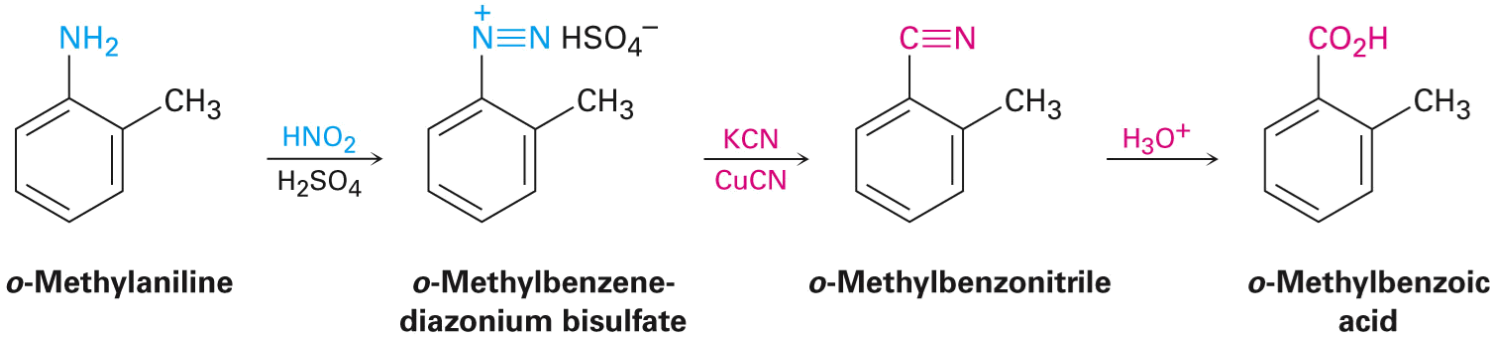
***p*-Methylaniline** **p-Bromotoluene (73%)**



Aniline **Iodobenzene (67%)**

Reactions of Arylamines

- An arenediazonium salt and CuCN yield the nitrile, ArCN, which can be hydrolyzed to other functional groups



o-Methylaniline

o-Methylbenzene-diazonium bisulfate

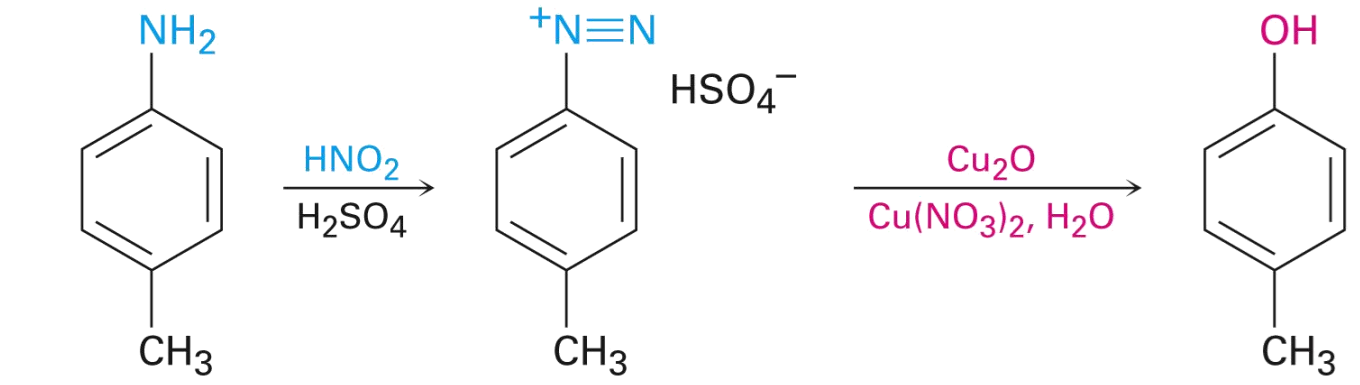
o-Methylbenzonitrile

o-Methylbenzoic acid

- The diazonio group be replaced by –OH to yield a phenol and by –H to yield an arene

Reactions of Arylamines

- Reaction of arenediazonium salt with copper(I) oxide in an aqueous solution of copper(II) nitrate yield phenols

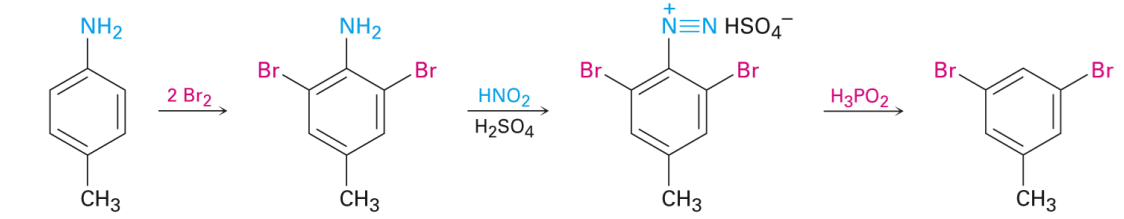


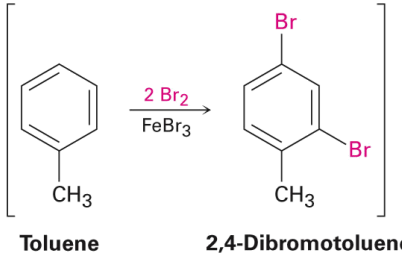
***p*-Methylaniline (*p*-Toluidine)**

***p*-Cresol (93%)**

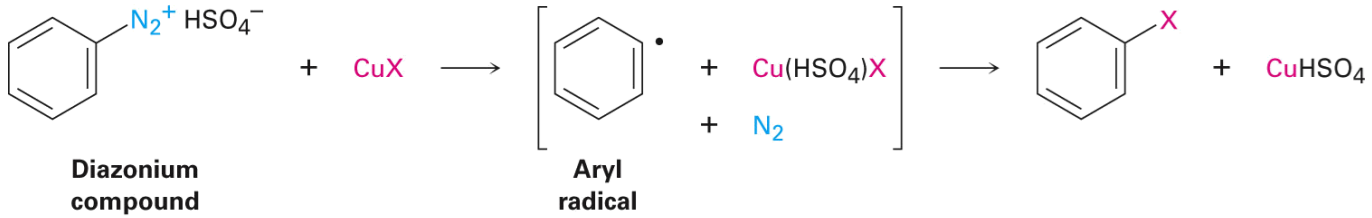
Reduction to a Hydrocarbon

- By treatment of a diazonium salt with hypophosphorous acid, H_3PO_2



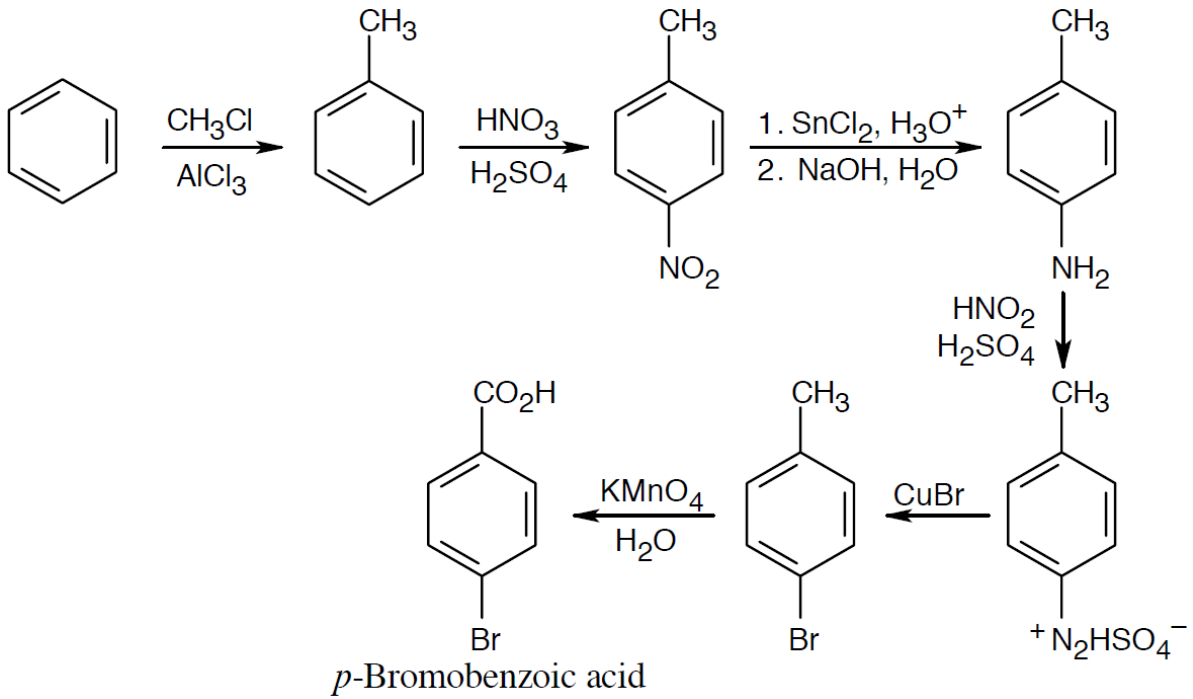


- Diazonium replacement takes place through radical pathways



Worked Example

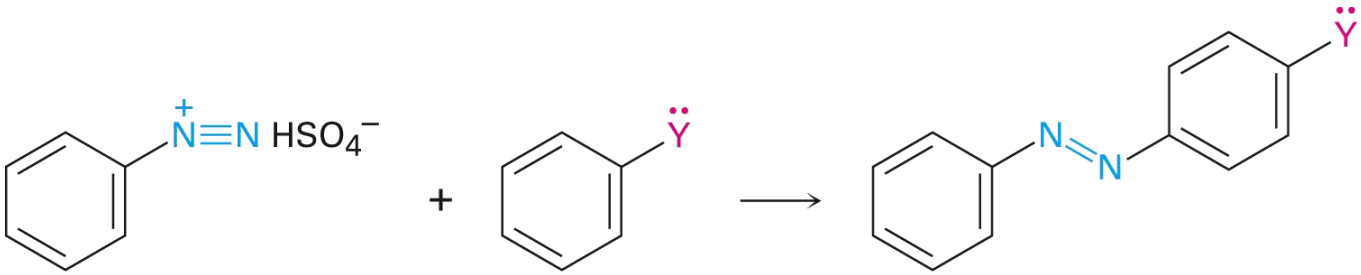
- How is *p*-bromobenzoic acid prepared from benzene using a diazonium replacement reaction



p-Bromobenzoic acid

Diazonium Coupling Reactions

- Arenediazonium salts undergo a coupling reaction with activated aromatic rings to yield **azo compounds**, Ar–N=N–Ar′



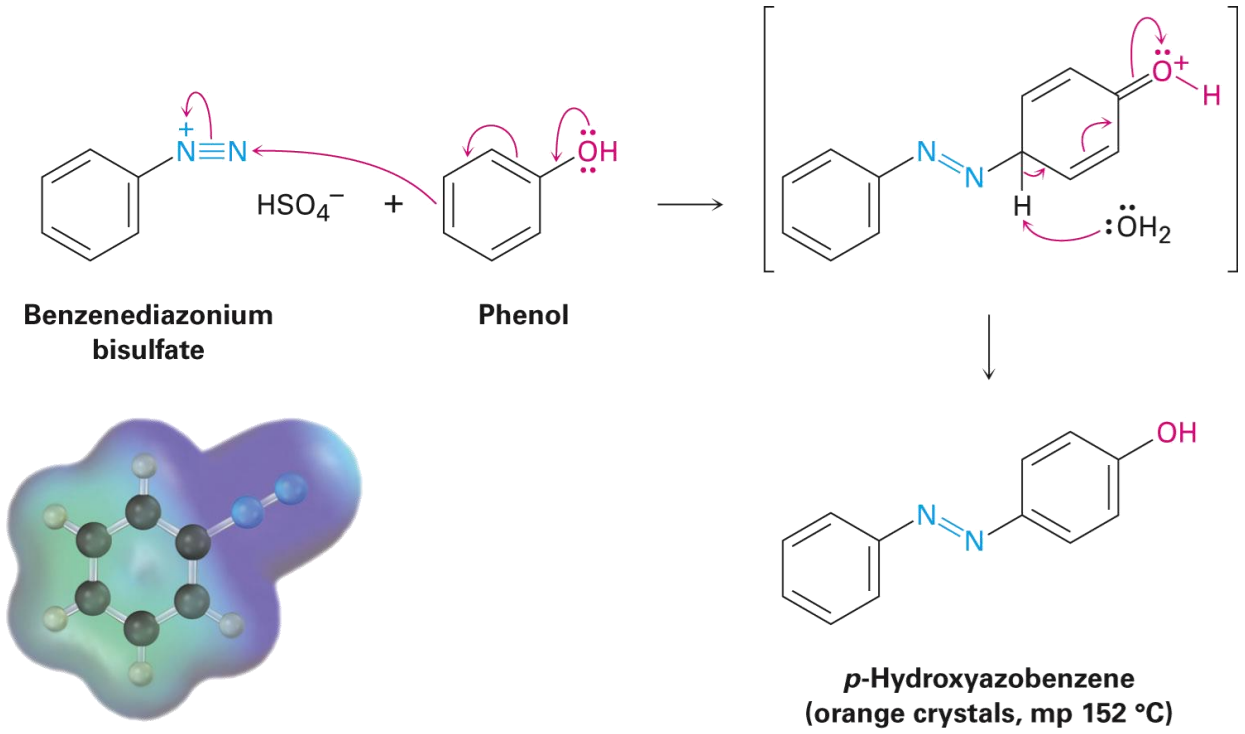
An azo compound

where Y = −OH or $-NR_2$

- Are typical electrophilic aromatic substitutions

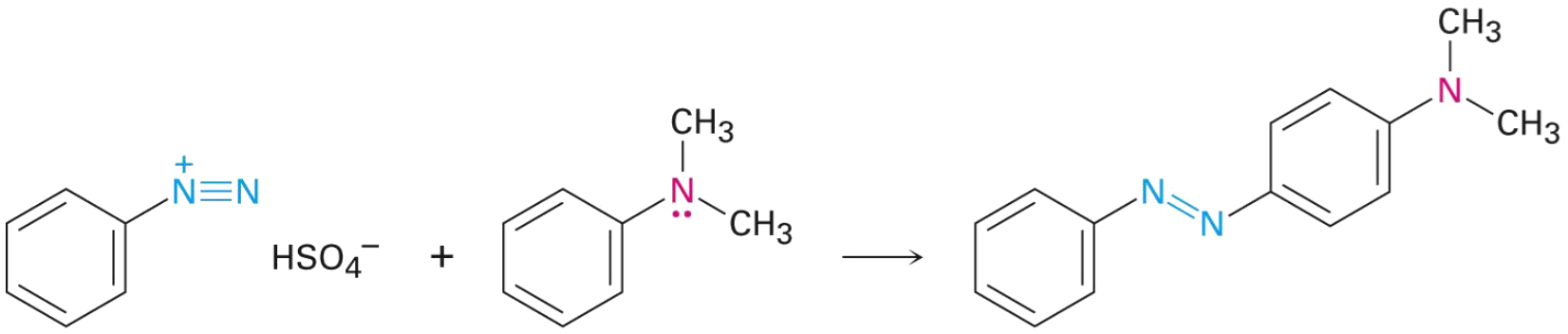
Diazonium Coupling Reactions

- The electrophilic diazonium ion reacts with the electron-rich ring of a phenol or arylamine
- Usually occurs at the para position but goes ortho if para is blocked



Diazonium Coupling Reactions

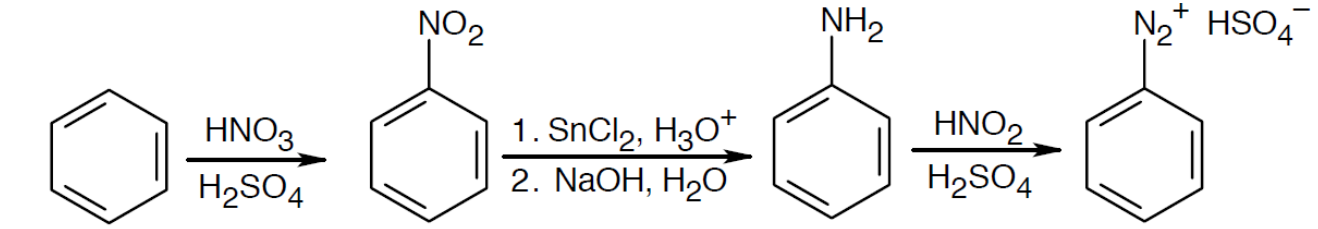
- Azo-coupled products have extended π conjugation that lead to low energy electronic transitions that occur in visible light

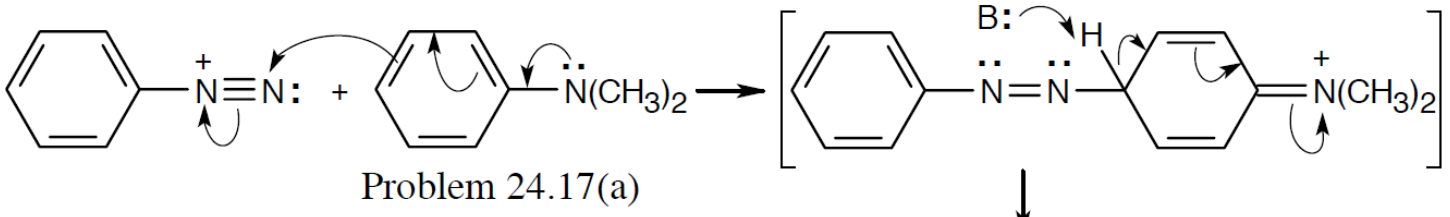


Benzenediazonium bisulfate + ***N,N*-Dimethylaniline** → ***p*-(Dimethylamino)azobenzene (yellow crystals, mp 127 °C)**

Worked Example

- Propose a synthesis of *p*-(dimethylamino)azobenzene with benzene as organic starting material



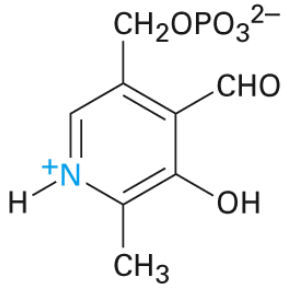


Problem 24.17(a)

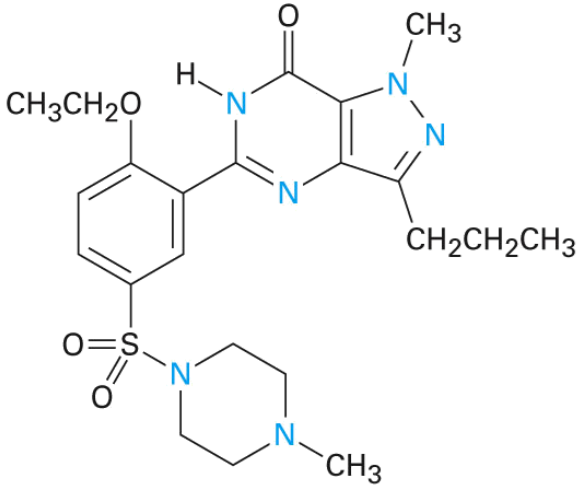
p-(*N*,*N*-Dimethylamino)azobenzene

Heterocyclic Amines

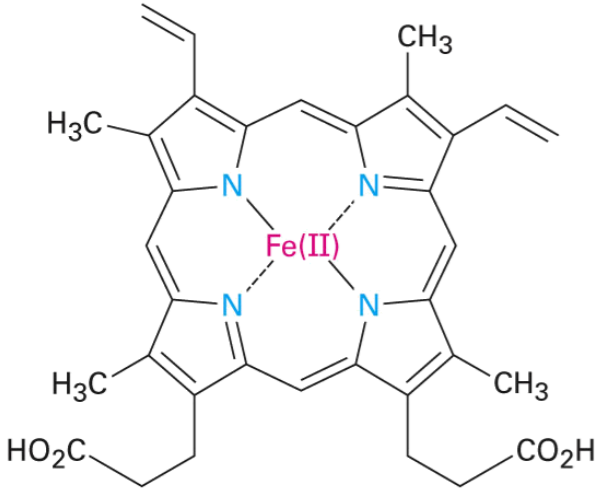
- A cyclic organic compound that contains atoms of two or more elements in its ring



Pyridoxal phosphate (a coenzyme)



Sildenafil (Viagra)

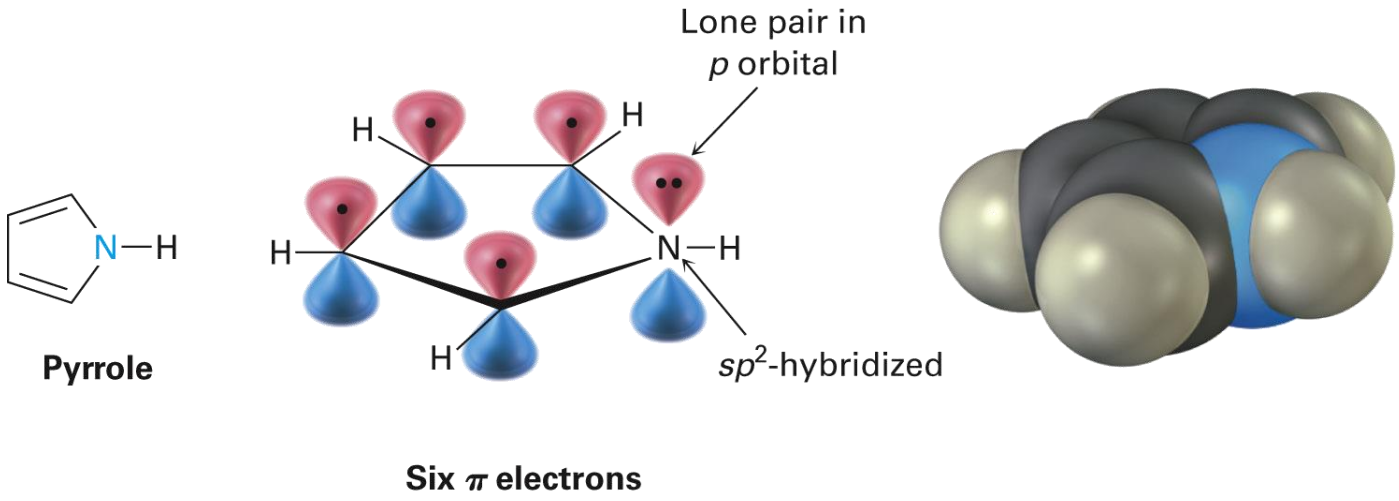


Heme

- Possess same chemistry as their open-chain counterparts

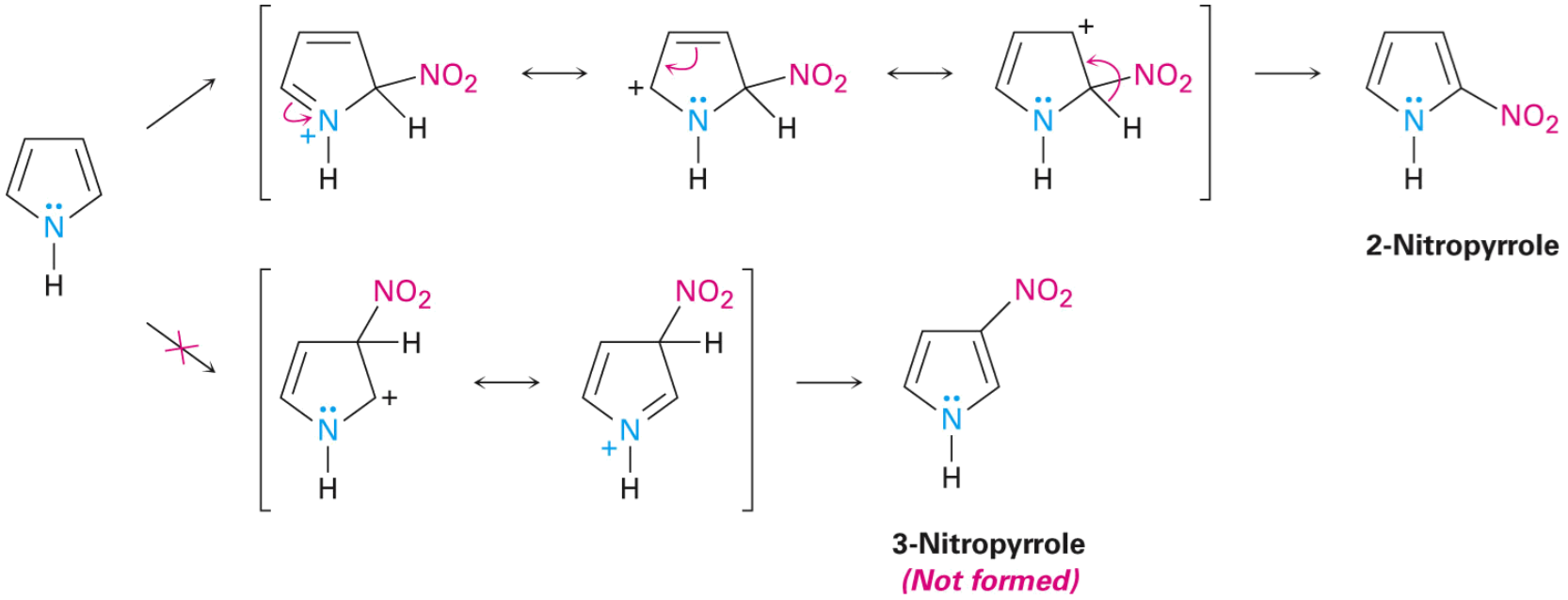
Heterocyclic Amines

- Pyrrole
  - Simplest five-membered unsaturated heterocyclic amine
    - Undergoes electrophilic substitution reactions
    - Chemical properties are not consistent with the structural features of amine or a conjugated diene



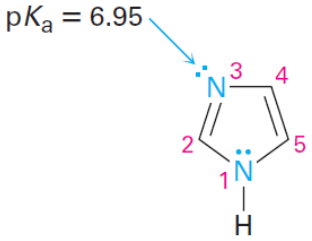
Heterocyclic Amines

- Ring is reactive toward electrophiles
- Electrophilic substitutions occur at C2
  - Reaction leads to intermediate cation having 3 resonance forms
- Reaction at C3 gives 2 resonance forms

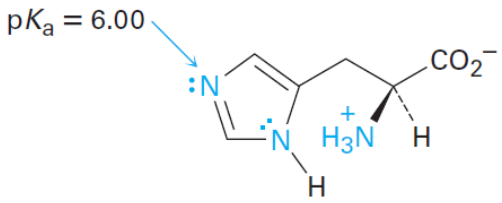


Heterocyclic Amines

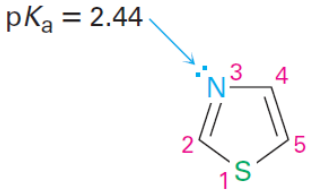
- Imidazole
  - Common five-membered heterocyclic amine
  - Constituent of the amino acid histidine
    - One of two nitrogens is basic



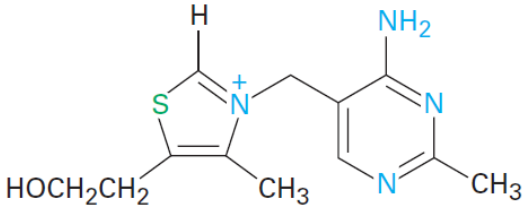
Imidazole



Histidine



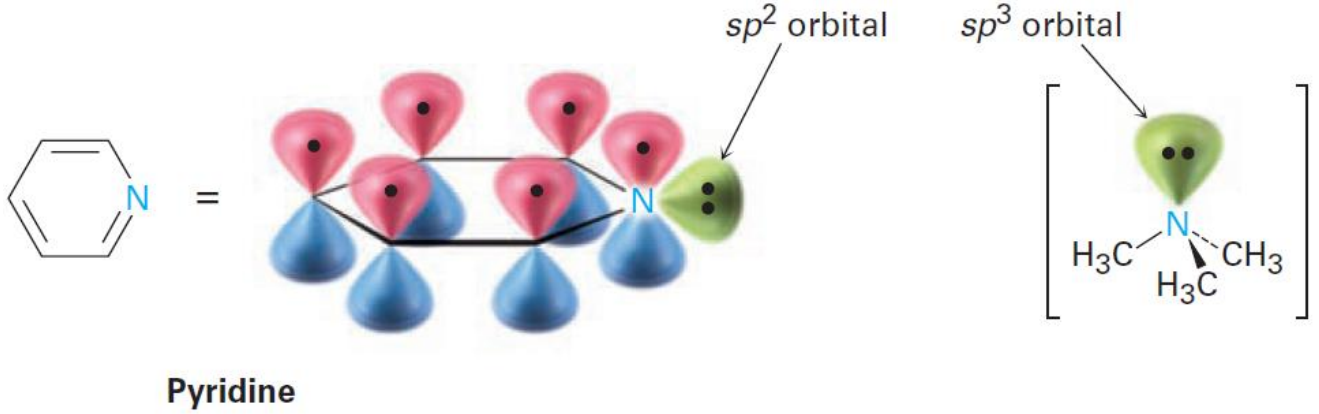
Thiazole



Thiamin (vitamin B_1)

Heterocyclic Amines

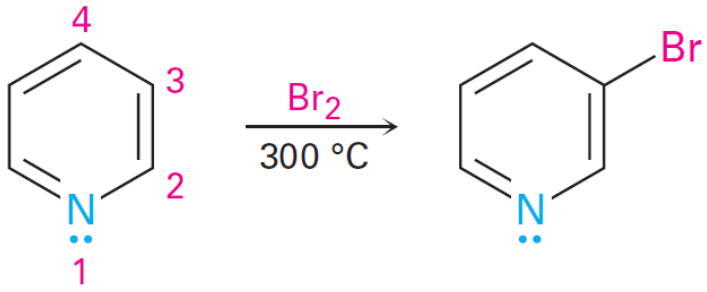
- Pyridine
  - Nitrogen-containing heterocyclic analog of benzene
    - The sp^2-hybridized nitrogen atom is less basic than the sp^3-hybridized nitrogen in an alkylamine



Pyridine

Heterocyclic Amines

- Undergoes electrophilic aromatic substitution reactions with difficulty

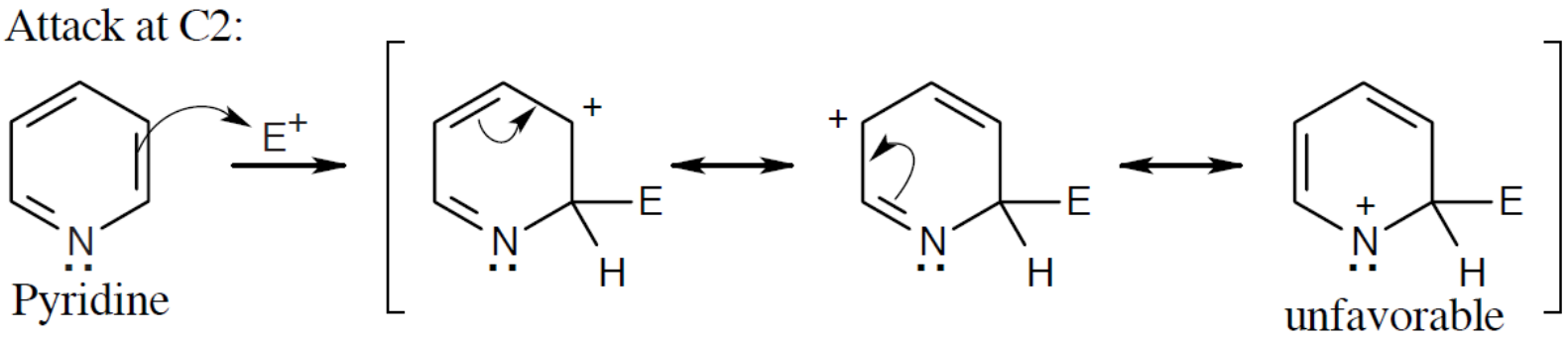


Pyridine **3-Bromopyridine (30%)**

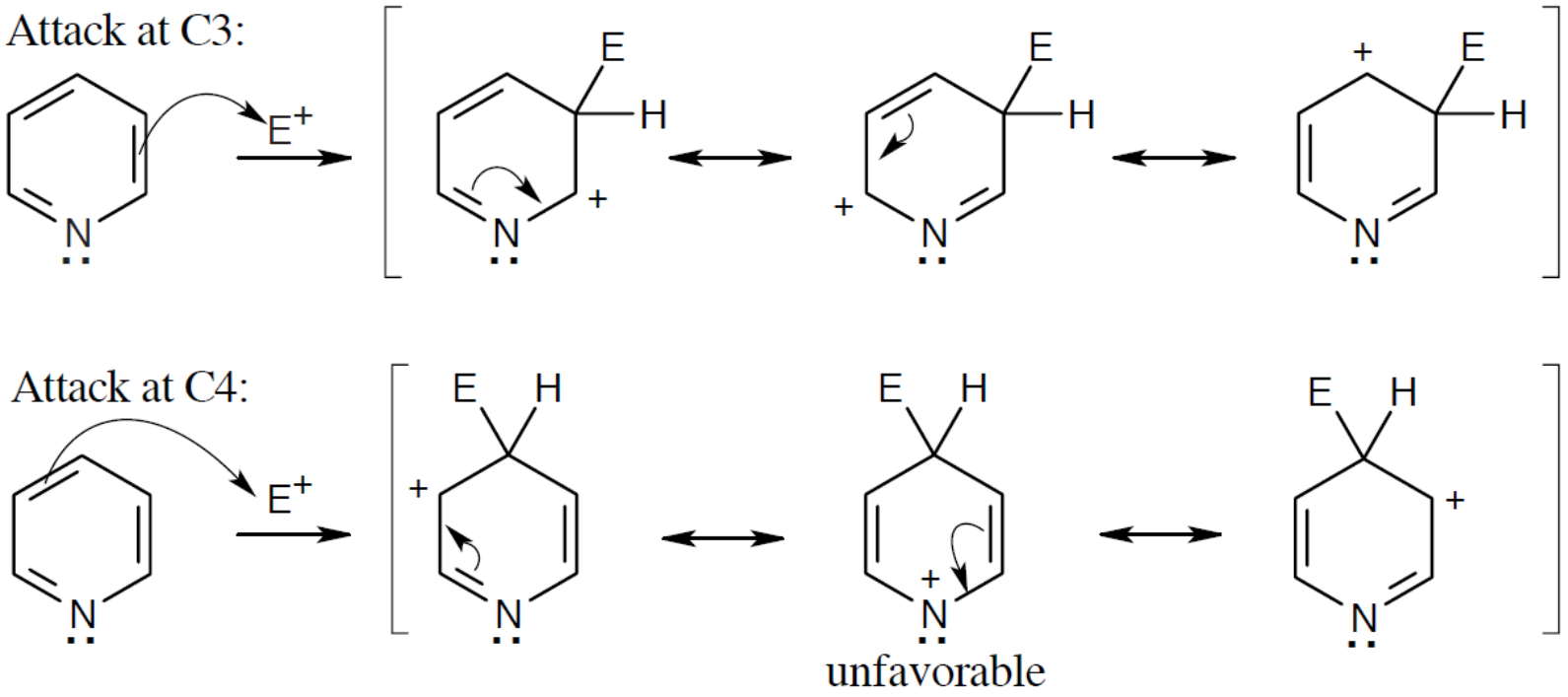
  - Acid-base complexation between the basic ring's nitrogen atom and the incoming electrophile
  - High dipole moment
- Six-membered diamine pyrimidine is found commonly in biological molecules

Worked Example

- Electrophilic aromatic substitution reactions of pyridine normally occur at C3
  - Draw the carbocation intermediates resulting from reaction of an electrophile at C2, C3, and C4
  - Explain the observed result
- Solution:

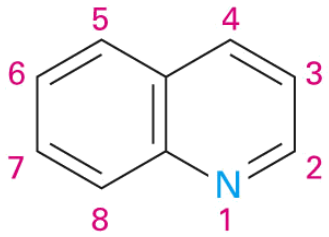


Worked Example

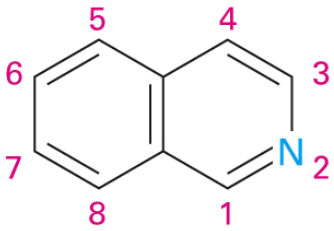


- Reaction at C3 is favored over reaction at C2 or C4

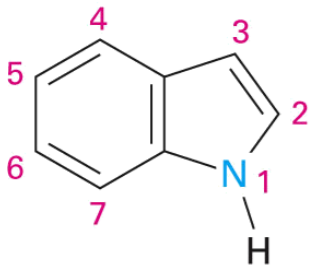
Heterocyclic Amines



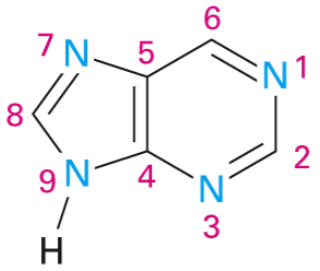
Quinoline



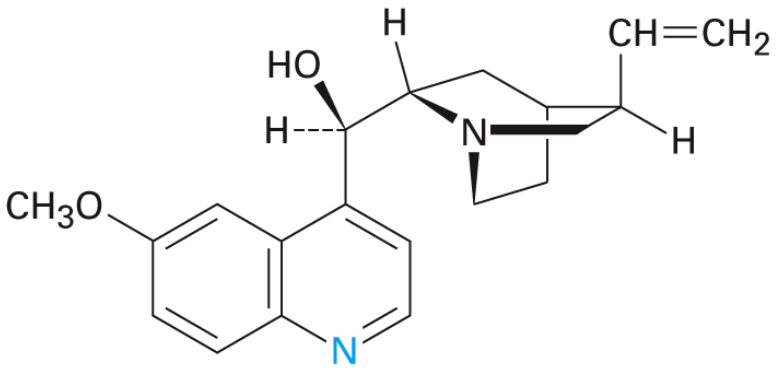
Isoquinoline



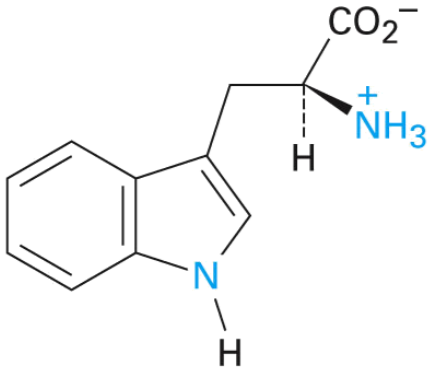
Indole



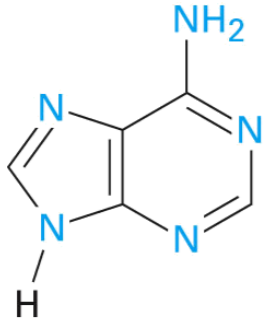
Purine



Quinine
(antimalarial)



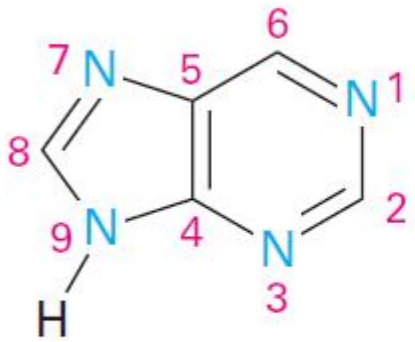
Tryptophan
(amino acid)

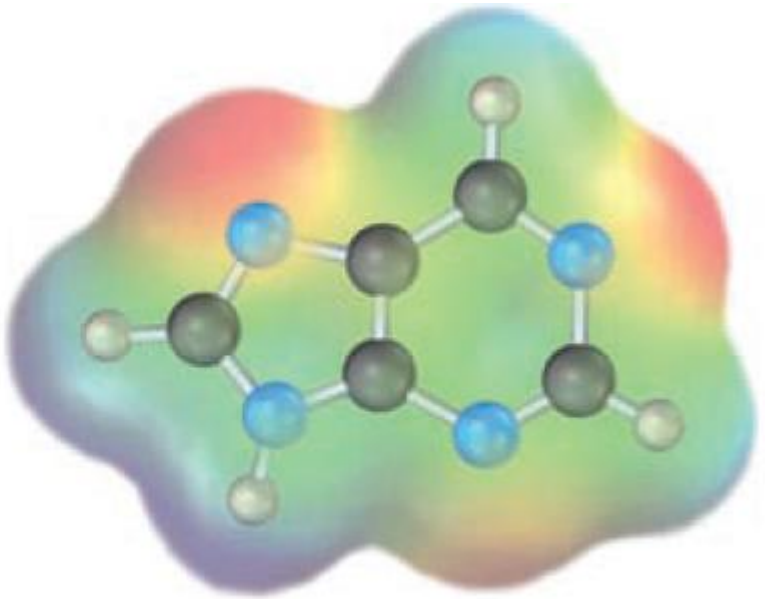


Adenine
(DNA constituent)

Heterocyclic Amines

- Purine has three basic, pyridine-like nitrogen atoms

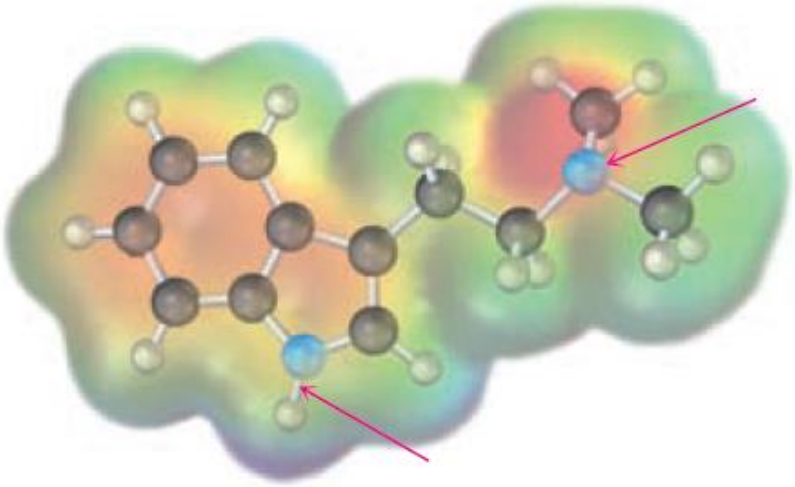




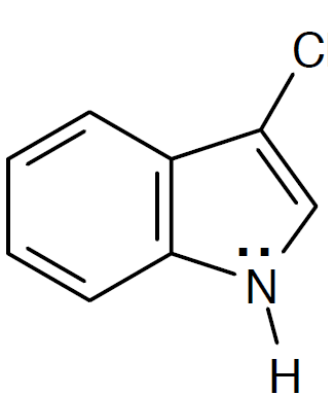
Purine

Worked Example

- Which nitrogen atom in the hallucinogenic indole alkaloid *N*,*N*-dimethyltryptamine is more basic?
  - Explain



***N*,*N*-Dimethyltryptamine**



N,*N*-Dimethyltryptamine